

SATakror

101 most repetitive questions of the Digital SAT



Lunar Intensity v.1

Find all 8 versions of this question at https://t.me/SATakror

Burt Kotler and colleagues showed that high moonlight intensity inhibits the activity of the greater Egyptian gerbil (*Gerbillus pyramidum*), a finding explicable in terms of benefits and costs: greater lunar intensity may not enable the gerbils to increase foraging success enough to offset the higher chance of detection by predatory foxes or owls. Though many other nocturnal mammals respond to lunar intensity variations similarly to greater Egyptian gerbils, Mohol bushbabies (*Galago moholi*) display the opposite pattern, as their heavy reliance on visual foraging results in a different balance of reward and risk.

1) Which choice best describes the function of the reference to "heavy reliance on visual foraging" in the text as a whole?

A) It emphasizes an attribute of Mohol bushbabies that enables them to respond to lunar intensity variations more rapidly than greater Egyptian gerbils and many other nocturnal mammals do.

B) It identifies a characteristic of Mohol bushbabies that explains why they do not reduce their activity under high lunar intensity like greater Egyptian gerbils and many other nocturnal mammals do.

C) It describes a behavioral pattern displayed by Mohol bushbabies that is inconsistent with how they would be expected to respond to high lunar intensity based on benefits and costs.

D) It presents a trait of Mohol bushbabies that serves as evidence for the assertion that greater Egyptian gerbils are not the only nocturnal mammals whose activity is inhibited by high lunar intensity.

Lunar Intensity v.2

**Find all 8 versions of this question at https://t.me/SATakror*

Optimal foraging theory (OFT) holds that animals' foraging behaviors reflect cost-benefit trade-offs that vary by species and with dynamic ecological circumstances. One such circumstance is lunar intensity, which Joel Brown and colleagues found to be negatively associated with foraging by Arizona pocket mice but Patricia C. Wright found to be positively associated with foraging by three-striped night monkeys. This discrepancy is explicable in terms of OFT: the monkeys' greater reliance on vision means that higher lunar intensity benefits them more than it benefits the mice.

2) Which choice best describes the finding made by Wright, as presented in the text?

A) During periods of high lunar intensity, three-striped night monkeys increase their reliance on vision.

B) As lunar intensity increases, Arizona pocket mice reduce their foraging activity.

C) As lunar intensity increases, three-striped night monkeys increase their foraging activity.

D) Increased lunar intensity is more beneficial for three-striped night monkeys than it is for Arizona pocket mice.

Lunar Intensity v.3

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Joel Brown and colleagues showed that high moonlight intensity inhibits the activity of the Arizona pocket mouse (Perognathus amplus), a finding explicable in terms of benefits and costs: greater lunar intensity may not enable the mice to increase foraging success enough to offset the higher chance of detection by predatory owls or hawks. Though many other nocturnal mammals respond to lunar intensity variations similarly to Arizona pocket mice, Azara's night monkeys (Aotus azarae) display the opposite pattern, as their heavy reliance on visual foraging results in a different balance of reward and risk.

3) Based on the text, why do Azara's night monkeys respond to high lunar intensity differently than Arizona pocket mice do?

A) In such conditions, the chance of detection by predators declines for Azara's night monkeys but rises for Arizona pocket mice.

B) Although the risks of activity in such conditions are greater for Azara's night monkeys than they are for Arizona pocket mice, the rewards are also greater for Azara's night monkeys.

C) In such conditions, the costs of activity are lower relative to the benefits of activity for Azara's night monkeys than they are for Arizona pocket mice.

D) Due to their heavy reliance on visual foraging, Azara's night monkeys experience a reduction in their foraging success in such conditions, whereas Arizona pocket mice experience no change in their foraging success.

Lunar Intensity v.4

**Find all 8 versions of this question at https://t.me/SATakror*

Optimal foraging theory (OFT) holds that animals' foraging behaviors reflect cost-benefit trade-offs that vary by species and with dynamic ecological circumstances. One such circumstance is lunar intensity, which Rodrigo A. Vásquez found to be negatively associated with foraging by olive grass mice but Deborah J. Curtis and colleagues found to be positively associated with foraging by mongoose lemurs. This discrepancy is explicable in terms of OFT: the lemurs' greater reliance on vision means that higher lunar intensity benefits them more than it benefits the mice.

4) Information in the text best supports which statement about OFT?

A) It tends to allow for a better understanding of the benefits of ecological circumstances than the costs of those circumstances.

B) It can explain why some species act in accordance with cost-benefit trade-offs and others do not.

C) It can account for observations of different species responding differently to similar ecological circumstances.

D) It may be weakened by the finding that the costs and benefits associated with a particular ecological circumstance vary by species.

Lunar Intensity v.5

*Find all 8 versions of this question at https://t.me/SATakror

Optimal foraging theory (OFT) holds that animals' foraging behaviors reflect cost-benefit trade-offs that vary by species and with dynamic ecological circumstances. One such circumstance is lunar intensity, which Mary V. Price and colleagues found to be negatively associated with foraging by white-throated woodrats but Simon Kenneth Bearder and colleagues found to be positively associated with foraging by Mohol bushbabies. This discrepancy is explicable in terms of OFT: the bushbabies' greater reliance on vision means that higher lunar intensity benefits them more than it benefits the woodrats.

5) Information in the text best supports which statement about Mohol bushbabies?

A) If increased lunar intensity imposed the same costs on the bushbabies that it imposes on white-throated woodrats, there would be no association between lunar intensity and the bushbabies' foraging.

B) If the advantages that the bushbabies gain from increased lunar intensity explain the change in their foraging behavior, those advantages are likely shared by some other species that are not heavily reliant on vision.

C) If increased lunar intensity creates any disadvantages for the bushbabies, those disadvantages are more than compensated for by the advantages that the bushbabies gain.

D) If the bushbabies' foraging behavior under increased lunar intensity actually reflects a cost-benefit trade-off, their behavior should be more similar to that of white-throated woodrats than it is.

Lunar Intensity v.6

**Find all 8 versions of this question at https://t.me/SATakror*

Vadamali Elangovan and Ganapathi Marimuthu showed that high moonlight intensity inhibits the activity of the greater short-nosed fruit bat (Cynopterus sphinx), a finding explicable in terms of benefits and costs: greater lunar intensity may not enable the bats to increase foraging success enough to offset the higher chance of detection by predatory owls or hawks. Though many other nocturnal mammals respond to lunar intensity variations similarly to greater short-nosed fruit bats, mongoose lemurs (Eulemur mongoz) display the opposite pattern, as their heavy reliance on visual foraging results in a different balance of reward and risk.

6) According to the text, what did Elangovan and Marimuthu find and why does that occur?

A) Greater short-nosed fruit bats tend to be more active during periods of high lunar intensity than at other times because such conditions allow them to increase their foraging success without making them easier to detect.

B) Greater short-nosed fruit bats reduce their activity during periods of high lunar intensity because predators can more easily spot the bats in brighter conditions, and such conditions do not benefit the bats enough to justify that risk.

C) During periods of high lunar intensity, greater short-nosed fruit bats reduce their activity because it is easier for predators to detect the bats in relatively bright conditions than it is for predators to detect mongoose lemurs in such conditions.

D) During periods of high lunar intensity, mongoose lemurs show a different behavioral response than greater short-nosed fruit bats and many other nocturnal mammals do because the risks to mongoose lemurs under such conditions are greater.

Lunar Intensity v.7

**Find all 8 versions of this question at https://t.me/SATakror*

Optimal foraging theory (OFT) holds that animals' foraging behaviors reflect cost-benefit trade-offs that vary by species and with dynamic ecological circumstances. One such circumstance is lunar intensity, which Rodrigo A. Vasquez found to be negatively associated with foraging by olive grass mice but Deborah J. Curtis and colleagues found to be positively associated with foraging by mongoose lemurs. This discrepancy is explicable in terms of OFT: the lemurs' greater reliance on vision means that higher lunar intensity benefits them more than it benefits the mice.

7) Information in the text best supports which statement about OFT?

A) It tends to allow for a better understanding of the benefits of ecological circumstances than the costs of those circumstances.

B) It can explain why some species act in accordance with cost-benefit trade-offs and others do not.

C) It can account for observations of different species responding differently to similar ecological circumstances.

D) It may be weakened by the finding that the costs and benefits associated with a particular ecological circumstance vary by species.

Lunar Intensity, v.8

**Find all 8 versions of this question at https://t.me/SATakror*

Vadamalai Elangovan and Ganapathi Marimuthu showed that high moonlight intensity inhibits the activity of the greater short-nosed fruit bat (Cynopterus sphinx), a result explicable in terms of benefits and costs: greater lunar intensity may not enable the bats to increase foraging success enough to offset the higher chance of detection by predatory owls or hawks. Most other nocturnal mammals respond to lunar intensity variations similarly to greater short-nosed fruit bats, but mongoose lemurs (Eulemur mongoz) display the opposite pattern, as their heavy reliance on visual foraging results in a different balance of reward and risk.

8) Which choice best describes the overall structure of the text?

A) It discusses two different responses to the same natural phenomenon, explains why one of those responses occurs, and then suggests that the other response still needs to be explained.

B) It describes and accounts for a finding, characterizes the finding as representative of a general pattern, and then describes and accounts for an exception to that pattern.

C) It presents and explains a study result, indicates that the result is similar to the results of many other studies, and then attributes a conflicting study result to a difference in that study's methods.

D) It introduces an observation of a behavioral pattern, presents an explanation for the pattern, and then describes an exception to the pattern that casts doubt on that explanation.

CMN, v.1

**Find all 4 versions of this question at https://t.me/SATakror*

Text 1

In separate studies, Marine Fernandez and colleagues and Xinhua He and colleagues examined whether plants transfer nutrients to one another using a common mycorrhizal network (CMN)—a lattice of fungal strands in the soil. Fernandez and colleagues excluded all pathways other than the CMN by using barriers to keep the plants' root systems separate while allowing mycorrhizal strands through—a crucial step He and colleagues' study did not take.

Text 2

Fernandez and colleagues took the necessary precaution of separating the plants' root systems (thereby excluding root-to-root transmission). However, any barrier used must allow the thread-like hyphae of a CMN to pass through, and this permeability would also allow liquids through. Thus, the researchers' experimental design cannot ensure that any nutrient transfer observed can be attributed to a CMN and not to some other pathway.

9) Based on the texts, which choice best describes a similarity in the points of view presented in Text 1 and Text 2?

A) Each text attempts to dispel a common misunderstanding about the likelihood of plant-to-plant nutrient transfer.

B) Each text assumes that most nutrient transfer between plants is via a CMN.

C) Each text analyzes methods for studying CMN nutrient transfer in order to propose an alternative method of study.

D) Each text critiques the methodology of a study about nutrient transfer via a CMN.

CMN, v.2

**Find all 4 versions of this question at https://t.me/SATakror*

Text 1

In separate studies, Stephen Meding and Robert J. Zasoski and Xinhua He and colleagues examined whether plants transfer nutrients to one another using a common mycorrhizal network (CMN)—a lattice of fungal strands in the soil. Meding and Zasoski excluded all pathways other than the CMN by using barriers to keep the plants' root systems separate while allowing mycorrhizal strands through—a crucial step He and colleagues' study did not take.

Text 2

Meding and Zasoski took the necessary precaution of separating the plants' root systems (thereby excluding root-to-root transmission). However, any barrier used must allow the thread-like hyphae of a CMN to pass through, and this permeability would also allow liquids through. Thus, the researchers' experimental design cannot ensure that any nutrient transfer observed can be attributed to a CMN and not to some other pathway.

10) Based on the texts, which choice best describes a point on which the author of Text 1 and the author of Text 2 would most likely agree?

A) Meding and Zasoski's study effectively excluded any explanation for nutrient transfer other than via a CMN.

B) Excluding root-to-root transfer of nutrients between plants is sufficient to ensure that any observed nutrient transfer must involve a CMN.

C) He and colleagues' study was not designed in a way that would allow it to produce compelling evidence that nutrient transfer occurred via a CMN.

D) A barrier that is impervious to both roots and fungal strands is necessary to evaluate nutrient transfer via a CMN.

CMN, v.3

**Find all 4 versions of this question at https://t.me/SATakror*

Text 1

In separate studies, Lingbo Meng and colleagues and Xinhua He and colleagues examined whether plants transfer nutrients to one another using a common mycorrhizal network (CMN)—a lattice of fungal strands in the soil. Meng and colleagues excluded all pathways other than the CMN by using barriers to keep the plants' root systems separate while allowing mycorrhizal strands through—a crucial step He and colleagues' study did not take.

Text 2

Meng and colleagues took the necessary precaution of separating the plants' root systems (thereby excluding root-to-root transmission). However, any barrier used must allow the thread-like hyphae of a CMN to pass through, and this permeability would also allow liquids through. Thus, the researchers' experimental design cannot ensure that any nutrient transfer observed can be attributed to a CMN and not to some other pathway.

11) Based on the texts, the author of Text 1 and the author of Text 2 would most likely give the same answer to which question?

A) Did He and colleagues' study effectively exclude any nutrient pathway other than a CMN?

B) Are the barriers used in Meng and colleagues' study sufficient to ensure that nutrient transfer could only occur via a CMN?

C) Is there an available barrier material that can block roots and liquids while allowing fungal strands through?

D) Do plants that transfer nutrients through a root-to-root pathway also transmit nutrients via a CMN?

CMN, v.4

**Find all 4 versions of this question at https://t.me/SATakror*

Text 1
In separate studies, Lingbo Meng and colleagues and Sanda Wahbi and colleagues examined whether plants transfer nutrients to one another using a common mycorrhizal network (CMN) —a lattice of fungal strands in the soil. Meng and colleagues excluded all pathways other than the CMN by using barriers to keep the plants' root systems separate while allowing mycorrhizal strands through —a crucial step Wahbi and colleagues' study did not take.

Text 2
Meng and colleagues took the necessary precaution of separating the plants' root systems (thereby excluding root-to-root transmission). However, any barrier used must allow the thread-like hyphae of a CMN to pass through, and this permeability would also allow liquids through. Thus, the researchers' experimental design cannot ensure that any nutrient transfer observed can be attributed to a CMN and not to some other pathway.

12) Based on the texts, the author of Text 1 and the author of Text 2 would most likely agree with which statement?

A) A barrier that is impervious to both roots and fungal strands is necessary to evaluate nutrient transfer via a CMN.

B) The barriers used in Meng and colleagues' study effectively excluded root-to-root transmission of nutrients.

C) Wahbi and colleagues' study did not find convincing evidence of nutrient transfer between individual plants.

D) Excluding root-to-root transfer of nutrients between plants is sufficient to ensure that any observed nutrient transfer must involve a CMN.

Sermon, v.1

The following text is adapted from Daniel Defoe's 1704 nonfiction book *The Storm.*

The sermon is a sound of words spoken to the ear, and prepared only for present meditation and extends no farther than the strength of memory can convey it; a book printed is a record, remaining in every man's possession, always ready to renew its acquaintance with his memory, and always ready to be produced as an authority or voucher to any reports he makes out of it, and conveys its contents for ages to come, to the eternity of mortal time, when the author is forgotten in his grave.

13) Which choice best states the main idea of the text?

A) People are less likely to forget a message when they hear it spoken aloud than they are when they read it in print.

B) Unless a spoken message is delivered by [a confident orator, it may be] ignored.

C) Most authors have little hope of being remembered well past their lifetimes.

D) Words committed to print have a greater permeance than messages that are merely spoken aloud.

Sermon, v.2

The following text is adapted from Daniel Defoe's 1704 nonfiction book *The storm.*

The sermon is a sound of words spoken to the ear, and prepared only for present meditation, and extends no farther than the strength of memory can convey it; a book printed is a record; remaining in every man's possession, always ready to renew its acquaintances with his memory, and always ready to be produced as an authority or voucher to any reports he makes out of it, and conveys its contents for ages to come, to the eternity of mortal time, when the author is forgotten in his grave.

14) Which choice best states the main purpose of the text?

A) It recounts why the works of some artists are still discussed by critics after the artists' deaths while others are not.

B) It analyzes the contributions to society of two different groups of scholars.

C) It demonstrates that one form of communication is more commonly used than another.

D) It describes differences in the characteristics of two kinds of communication.

Arthurian Legends, v.1

**Find all 4 versions of this question at https://t.me/SATakror*

Arthurian legends (tales related to the character of King Arthur) derive from many sources, such as *Preiddeu Annwn*, composed around 900, and *Perceval, the Story of the Grail* from around 1181. Sir Thomas Malory's 15th-century text *Le Morte d'Arthur* was an attempt to compile these stories into a coherent narrative. Many of Malory's sources derive from *Geoffrey of Monmouth's History of the Kings of Britain*, written in the 1130s. While neither *History* nor any works that predate it mention Arthur's famous Round Table at which his knights assembled, *Le Morte d'Arthur* does, suggesting that _____

15) Which choice most logically completes the text?

A) Geoffrey of Monmouth's accounts of Arthurian legends in *History* are more similar overall in content to the accounts in *Perceval, the Story of the Grail* than they are to the account in *Le Morte d'Arthur*.

B) Malory encountered the Round Table in a version that Geoffrey of Monmouth was not familiar with when writing his *History*.

C) *Le Morte d'Arthur* is more historically accurate than *History*, because *Perceval, the Story of the Grail* had not been written when Geoffrey of Monmouth was writing his work.

D) When a version of an Arthurian legend contradicted the version in *History*, Malory preferred to include Geoffrey of Monmouth's version in *Le Morte d'Arthur*.

Arthurian Legends, v.2

**Find all 4 versions of this question at https://t.me/SATakror*

Arthurian legends (tales related to the character of King Arthur) derive from many sources, such as *Preiddeu Annwn*, composed around 900, and *The High Book of the Grail* from the early 13th century. One of the most significant sources, *Geoffrey of Monmouth's History of the Kings of Britain*, was written in Latin in the 1130s; some material from it was later adapted by the Norman poet Wace into the *Roman de Brut* in 1155. But while the *Roman de Brut* includes references to the famous Round Table at which Arthur's knights assembled, those seem to be Wace's own inventions, which suggests that _____

16) Which choice most logically completes the text?

A) Geoffrey of Monmouth's accounts of Arthurian legends in his *History* are more similar overall in content to the accounts in *The High Book of the Grail* than they are to the accounts in *Roman de Brut*.

B) no Arthurian legend that was available to Geoffrey of Monmouth had any mention of the Round Table.

C) Geoffrey of Monmouth was unaware of stories of the Round Table when composing his *History*, though historians know that works containing such stories were available to him.

D) *Roman de Brut* is more historically accurate than *History*, because *The High Book of the Grail* had not been written when Geoffrey of Monmouth was writing his work.

Arthurian Legends, v.3

**Find all 4 versions of this question at https://t.me/SATakror*

17

Arthurian legends (tales related to the character of King Arthur) derive from many sources, such as *Preiddeu Annwfn*, composed around 900, and *Erec and Enide* from around 1170. One of the most significant sources, Geoffrey of Monmouth's *History of the Kings of Britain*, was written in the 1130s; some material from it was later adapted by the Norman poet Wace into the *Roman de Brut* in 1155. But while the *Roman de Brut* includes references to the famous Round Table at which Arthur's knights assembled, nothing written before 1155 does, which suggests that _____

17) Which choice most logically completes the text?

A) *Roman de Brut* is more historically accurate than *History*, because *Erec and Enide* had not been written when Geoffrey of Monmouth was writing his work.

B) Geoffrey of Monmouth was unaware of stories of the Round Table when composing his *History*, though historians know that works containing such stories were available to him.

C) Geoffrey of Monmouth's accounts of Arthurian legends in his *History* are more similar overall in content to the accounts in *Erec and Enide* than they are to the accounts in *Roman de Brut*.

D) Any Arthurian legend that mentions the Round Table likely has a lineage that connects it to *Roman de Brut*.

Arthurian Legends, v.4

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Arthurian legends (tales related to the character of King Arthur) derive from many sources, such as *Preiddeu Annwfn*, composed around 900, and *Perceval, the Story of the Grail* from around 1181. Sir Thomas Malory's 15th-century text *Le Morte d'Arthur* was an attempt to compile these stories into a coherent narrative. Many of Malory's sources derive from Geoffrey of Monmouth's *History of the Kings of Britain*, written in the 1130s. While neither *History* nor any works that predate it mention Arthur's famous Round Table at which his knights assembled, *Le Morte d'Arthur* does, suggesting that _____

18) Which choice most logically completes the text?

A) Geoffrey of Monmouth's accounts of Arthurian legends in *History* are more similar overall in content to the accounts in *Perceval, the Story of the Grail* than they are to the account in *Le Morte d'Arthur*.

B) Malory encountered the Round Table in a version that Geoffrey of Monmouth was not familiar with when writing his *History*.

C) *Le Morte d'Arthur* is more historically accurate than *History*, because *Perceval, the Story of the Grail* had not been written when Geoffrey of Monmouth was writing his work.

D) When a version of an Arthurian legend contradicted the version in *History*, Malory preferred to include Geoffrey of Monmouth's version in *Le Morte d'Arthur*.

Chimpanzees and Sunlight, v.1

**Find all 3 versions of this question at https://t.me/SATakror*

Exclusively inhabiting tropical countries such as Ghana, wild chimpanzees lack adaptations to seasonal variations in ultraviolet B (UVB) irradiance from sunlight; since UVB exposure enables vertebrates to synthesize vitamin D, Sophie Moitié and colleagues studied zoo chimpanzees in Spain and other mid-latitude countries to see how vitamin D levels are affected by the seasonal variations in UVB irradiance that occur in those locations. They found that chimpanzees' vitamin D levels were significantly lower in autumn than in summer and appeared unaffected by oral supplementation of vitamin D administered by zookeepers. Moitié and colleagues point out, however, that supplementation was rare, highly varied, and poorly tracked, and therefore _____

19) Which choice most logically completes the text?

A) the effect of supplemental vitamin D on zoo chimpanzees in Spain and other mid-latitude countries can more clearly be observed in summer than in autumn.

B) the possibility that zoo chimpanzees in Spain and other mid-latitude countries would benefit from supplemental vitamin D during autumn cannot be excluded.

C) vitamin D levels in zoo chimpanzees in Spain and other mid-latitude countries may be higher during autumn than Moitié and colleagues' data appear to indicate.

D) differences in vitamin D supplementation likely explain more of the difference in vitamin D levels across zoo chimpanzees in Spain and other mid-latitude countries than seasonal differences in UVB irradiation do.

Chimpanzees and Sunlight, v.2

**Find all 3 versions of this question at https://t.me/SATakror*

Exclusively inhabiting tropical countries such as Liberia, wild chimpanzees lack adaptations to seasonal variations in ultraviolet B (UVB) irradiance from sunlight; since UVB exposure enables vertebrates to synthesize vitamin D, this raises questions about how chimpanzees in mid-latitude zoos are affected by the seasonal variations in UVB irradiance that occur in those locations. In a study of zoo chimpanzees in Belgium and other mid-latitude countries, Sophie Moitié and colleagues found that vitamin D levels are lower in spring than in summer, paralleling UVB irradiance, but also that regardless of season, chimpanzees with unlimited outdoor access have higher vitamin D levels than chimpanzees with limited outdoor access. These findings suggest that _____

20) Which choice most logically completes the text?

A) giving zoo chimpanzees in Belgium and other mid-latitude countries unlimited outdoor access during spring may help offset the effect of reduced UVB irradiance relative to summer.

B) adaptations to seasonal variations in UVB irradiance may be emerging in zoo chimpanzees with unlimited outdoor access in Belgium and other mid-latitude countries.

C) the amount of outdoor access received by zoo chimpanzees in Belgium and other mid-latitude countries affects those chimpanzees' vitamin D levels more than do seasonal variations in UVB irradiance.

D) in mid-latitude countries such as Belgium, zoo chimpanzees with unlimited outdoor access tend to synthesize more vitamin D per unit of UVB irradiance in summer than they do in spring.

Chimpanzees and Sunlight, v.3

**Find all 3 versions of this question at https://t.me/SATakror*

Exclusively inhabiting tropical countries such as Liberia, wild chimpanzees lack adaptations to seasonal variations in ultraviolet B (UVB) irradiance from sunlight; since UVB exposure enables vertebrates to synthesize vitamin D, this raises questions about how chimpanzees in mid-latitude zoos are affected by the lower and more variable UVB irradiance in those locations. In a study of zoo chimpanzees in Norway and other mid-latitude countries, Sophie Moitié and colleagues found not only that chimpanzees' vitamin D levels correlate with UVB irradiance but also that vitamin D levels show no evidence of plateauing as UVB irradiance reaches its highest local levels, suggesting that _____

21) Which choice most logically completes the text?

A) averaged across seasons, vitamin D levels in zoo chimpanzees in mid-latitude countries such as Norway tend to be comparable to those in wild chimpanzees in tropical countries such as Liberia.

B) adaptations to seasonal variations in UVB irradiance may be emerging in zoo chimpanzees in Norway and other mid-latitude countries.

C) zoo chimpanzees in Norway and other mid-latitude countries tend to synthesize less vitamin D than they are inherently capable of synthesizing.

D) providing supplemental vitamin D to chimpanzees in zoos in Norway and other mid-latitude countries would likely not be beneficial.

Types of Steel, v.1

**Find all 3 versions of this question at https://t.me/SATakror*

All stainless steel contains varying amounts of iron, carbon, and corrosion-inhibiting chromium. However, ferritic stainless steel, often used for cooking utensils, contains a higher percentage of chromium (at least 10.5%) than does austenitic stainless steel and a higher concentration of iron, which is responsible for its magnetic properties. Unlike ferritic stainless steel, austenitic stainless steel has a face-centered cubic crystalline structure resulting from the addition of nickel to the alloy. Austenitic stainless steel has two subtypes: the 200 series, often used for washing machines, and the 300 series, which has more nickel than the 200 series and is often used for mining and chemical equipment or compensators. Thus, stainless steel used to manufacture _____

22) Which choice most logically completes the text?

A) cooking utensils will have a face-centered cubic crystalline structure.

B) mining and chemical equipment will have a concentration of chromium lower than 10.5%, while stainless steel used for washing machines will not.

C) cooking utensils will have greater magnetic properties than stainless steel used to manufacture mining and chemical. equipment will

D) compensators will likely have a similar crystalline structure to stainless steel used to manufacture cooking utensils.

Types of Steel, v.2

**Find all 3 versions of this question at https://t.me/SATakror*

All stainless steel contains varying amounts of iron, carbon, and corrosion-inhibiting chromium. However, ferritic stainless steel, often used for induction cookers, contains a higher percentage of chromium (at least 10.5%) than does austenitic stainless steel and a higher concentration of iron, which is responsible for its magnetic properties. Unlike ferritic stainless steel, austenitic stainless steel has a face-centered cubic crystalline structure resulting from the addition of nickel to the alloy. Austenitic stainless steel has two subtypes: the 300 series, often used for storage containers, and the 200 series, which has less nickel and more nitrogen than the 300 series and is used for dishwashers. Thus, stainless steel used to manufacture _____

23) Which choice most logically completes the text?

A) both storage containers and dishwashers will have a face-centered cubic crystalline structure, but stainless steel used to manufacture storage containers will have less nickel than stainless steel used to manufacture dishwashers will.

B) storage containers will have less nitrogen in its composition than stainless steel used to manufacture dishwashers will.

C) storage containers will have a concentration of nitrogen greater than 10.5 %, while stainless steel used to manufacture dishwashers will not.

D) storage containers will have similar magnetic properties to stainless steel used to manufacture induction cookers.

Types of Steel, v.3

**Find all 3 versions of this question at https://t.me/SATakror*

All stainless steel contains varying amounts of iron, carbon, and corrosion-inhibiting chromium. However, ferritic stainless steel, often used for induction cookers, contains a higher percentage of chromium (at least 10.5%) than does austenitic stainless steel and a higher concentration of iron, which is responsible for its magnetic properties. Unlike ferritic stainless steel, austenitic stainless steel has a face-centered cubic crystalline structure resulting from the addition of nickel to the alloy. Austenitic stainless steel has two subtypes: the 300 series, often used for storage containers, and the 200 series, which has less nickel and more nitrogen than the 300 series and is used for home water tanks. Thus, stainless steel used to manufacture storage containers will have _____

24) Which choice most logically completes the text?

A) similar magnetic properties to stainless steel used to manufacture induction cookers.

B) lower percentage of nickel in its composition than stainless steel used to manufacture induction cookers will.

C) a face-centered cubic crystalline structure, while stainless steel used to manufacture induction cookers will not.

D) concentration of nitrogen greater than 10.5%, while stainless steel used to manufacture home water tanks will not.

Ecology in Indigenous songs, v.1

**Find all 5 versions of this question at https://t.me/SATakror*

Ethnographers have spent decades analyzing the traditional songs of Indigenous peoples around the world. Only recently, however, have ethnographers started to recognize that many Indigenous songs encode and transmit valuable ecological knowledge—citing, for instance, songs of the Kwakwaka'wakw people in British Columbia, Canada, that share vital information about clam gardens, and those of the Sakha people in Siberia, Russia, that address the effects of climate change on local ecosystems. It seems likely, therefore, that ethnographers may want to _____

25) Which choice most logically completes the text?

A) reconsider their view that the transmission of ecological knowledge is an important function of some Indigenous songs.

B) reexamine some of their earlier analyses of Indigenous peoples' songs in light of this new perspective.

C) explore the possibility that the Kwakwaka'wakw people were influenced by the songs of the Sakha.

D) place greater emphasis on the use of Indigenous songs by individuals than on the songs' roles in society.

Ecology in Indigenous songs, v.2

**Find all 5 versions of this question at https://t.me/SATakror*

Indigenous songs are an important resource for ecological information. Songs of the Warlpiri people in Yuendumu, Australia, convey information about wild edible seeds, and the songs of the Kaluli people in Papua New Guinea replicate rainforest sounds to evoke specific locations. There have been efforts made to preserve Indigenous languages over the years — e.g., the United Nations International Decade of Indigenous Languages (2022-2032). However, such attempts have typically focused on spoken language despite the fact that some expressions in these languages appear only in songs, Therefore, if those involved in such efforts want to ensure that a comprehensive range of information is secured, they must _____

26) Which choice most logically completes the text?

A) recognize that Indigenous languages likely have more unique expressions that represent ecological knowledge than represent other types of information.

B) incorporate the preservation of songs into their broader efforts to protect Indigenous languages.

C) analyze musical similarities in the songs of the Warlpiri and Kaluli people.

D) acknowledge that Indigenous songs are largely but not entirely composed using phrases taken from spoken language.

Ecology in Indigenous songs, v.3

**Find all 5 versions of this question at https://t.me/SATakror*

As exemplified by Temiar songs about landforms and landmarks and Lakota songs about gathering mouse beans, ecological information can be transmitted in Indigenous songs, and in some instances is maintained only in this way. Kwaxsistalla Wathl'thla, a song keeper for the Kwakwaka'wakw people in Canada, collaborated with ethnobiologist Dana Lepofsky et al., sharing songs referencing terraced intertidal clam gardens the people implemented in the past to foster healthy development of a dietary staple. Drawing on archaeological evidence as well, Lepofsky et al. determined that the prevalence of the practice described in the songs corresponded with growth in clam size and abundance despite increased harvesting pressure—a finding that demonstrates that _____

27) Which choice most logically completes the text?

A) representation of practical applications of ecological knowledge is the defining characteristic of the music of certain Indigenous peoples.

B) the Kwakwaka'wakw people likely would not have detailed their creation of clam gardens in songs if their efforts had not produced significantly larger clams.

C) the clams harvested from intertidal terraces by Kwakwaka'wakw people in the past likely were a different species than the clams found in those areas today.

D) effective methods for the cultivation of sources of sustenance are among the ecological knowledge preserved in Indigenous songs.

Ecology in Indigenous songs, v.4

**Find all 5 versions of this question at https://t.me/SATakror*

Scholars are increasingly exploring the communication and preservation of ecological knowledge through Indigenous songs (e.g., Temiar songs about landforms and landmarks and those of the Tlingit people about wildlife encounters). In one study, ethnobiologist Dana Lepofsky et al. received insight from Kwaxsistalla Wathl'thla, a song keeper for the Kwakwaka'wakw people in Canada, into songs referencing the people's use of terraced gardens in intertidal zones along the Pacific Northwest coast for the cultivation of clams for consumption. Archaeological evidence of significant increases in clam size and abundance in that area concurrent with the documented past implementation of the method described in the songs supports the conclusion that _____

28) Which choice most logically completes the text?

A) there is greater corroboration in the archaeological record of ecological practices described in Kwakwaka'wakw songs than of those described in Temiar and Tlingit songs.

B) non-Indigenous people around the Pacific Northwest coast adopted the practice developed by the Kwakwaka'wakw people after observing its efficacy.

C) although contemporary Kwakwaka'wakw people have a deep understanding of and appreciation for the fishing and farming practices used by their ancestors, they no longer implement those methods.

D) the practice used by ancestors of modern Kwakwaka'wakw people not only effectively maintained a food source but also promoted its robustness.

Ecology in Indigenous songs, v.5

**Find all 5 versions of this question at https://t.me/SATakror*

Indigenous songs can be repositories of ecological information, from Yi songs about the natural environment to Tlingit songs about wildlife encounters. Kwakwakawakw song keeper Kwaxsistalla Wathlthla aided research by ethnobiologist Dana Lepofsky et al. by contributing songs that reference creating terraced intertidal gardens to foster the healthy development of clams, a staple in the Kwakwakawakw people's diet. Such structures date as far back as 3,500 years ago. Intriguingly, analysis of local paleoecological marine bivalve records has revealed significant increases in clam size and productivity as clam garden practices became more prevalent and harvesting pressure increased—a finding that bolsters the notion that _____

29) Which choice most logically completes the text?

A) the Kwakwakawakw people encouraged the health of clam gardens and reduced pressure on them by periodically increasing their reliance on other sources of sustenance.

B) the practical approach described in the Kwakwakawakw songs effectively maintained clams as a sustainable food source.

C) the Kwakwakawakw people initially adapted their clam garden practices from similar methods observed among other Indigenous peoples in neighboring areas.

D) contemporary Kwakwakawakw people have only minimally deviated from the clam-gardening practices described in the people's songs.

Brand Extension, v.1

**Find all 3 versions of this question at https://t.me/SATakror*

The Virgin Group's introduction of the Virgin Atlantic airline in 1984 is an instance of brand extension—the company leveraged its brand recognition as a music retailer to enter a new product category. To investigate how market share affects consumers' likelihood of purchasing brand extensions, Alicia Grasby et al. identified 30 extended brand pairs (e.g., the same brand of vacuum cleaner and microwave) in household purchasing data; for each pair, Grasby et al. calculated the branded products' market share and calculated the increase in probability of a brand in one category being purchased if the same brand was purchased in the other category. A broad inverse relationship between the two values emerged, which can most reasonably be attributed to the fact that _____

30) Which choice most logically completes the text?

A) the purchase probability of a brand is affected by the purchase probability of the particular product category to which it belongs, and thus brands in categories that are rarely purchased tend to have low purchase probabilities regardless of market share.

B) as a brand's market share increases, the number of competing brands in the same product category tends to decrease, and thus the rate of increase in market share tends to accelerate as a brand's market share grows.

C) brands with high market share have high purchase probability regardless of whether the other product in the pair is purchased and thus have less potential to increase in purchase probability than brands with low market share do.

D) consumers tend to be less familiar with brands with low market share than brands with high market share and thus may purchase both products in a pair with low market share without recognizing that the products are the same brand.

Brand Extension, v.2

**Find all 3 versions of this question at https://t.me/SATakror*

Apple's introduction of the iPhone in 2007 is a quintessential instance of brand extension— the company leveraged its brand recognition as a computer manufacturer to enter a product category where it had not previously competed. An outstanding question is whether perceived category similarity predicts consumers' likelihood of purchasing brand extensions. To answer this question, Alicia Grasby et al. identified 30 extended-brand pairs (e.g., the same brand of laundry detergent and air freshener) in 52 weeks of purchases by approximately 60,000 households and, for each pair, calculated the change in probability of a brand in one category being purchased if the same brand was purchased in the other category.

31) Based on the text, which potential study design would be most likely to produce evidence that would enable Grasby et al. to answer their research question?

A) Have a representative sample of the households rate the similarity of the product categories in each extended-brand pair, then determine how, if at all, those ratings correlate with the change in probability that the team calculated for each pair.

B) Have a representative sample of the households rate the similarity of one product in each extended-brand pair to other products in the same category, then determine how, if at all, those ratings correlate with the change in probability that the team calculated for each pair.

C) Poll a representative sample of the households to determine the degree of brand recognition for each brand in the extended-brand pairs, then determine how, if at all, the degree of brand recognition correlates with the frequency with which a different group of households purchased at least one product of that brand.

D) Poll a representative sample of the households to determine the degree of brand recognition of each brand in the extended-brand pairs, then determine how, if at all, the degree of brand recognition correlates with the average cost of each product in the pairs.

Brand Extension, v.3

**Find all 3 versions of this question at https://t.me/SATakror*

Amazon's introduction of the AWS cloud-computing platform in 2002 is an instance of brand extension—the company leveraged its brand recognition as an online retailer to enter a new product category. To understand consumer responses to extended brands, Alicia Grasby et al. identified 30 extended-brand pairs (e.g., the brand of automobile and lawnmower) in household purchasing data and, for each pair, had consumers rate the similarity of the pair's product categories. Pairs of product categories such as bread and bagels were rated as similar, whereas those such as mozzarella cheese and ice cream were rated as dissimilar. Grasby et al. concluded that category similarity positively affects the probability of cross-category purchases of the same brand.

32) Which finding, if true, would most directly support the conclusion put forward by Grasby et al.?

A) Extended-brand pairs with high product-category similarity ratings tended to be in categories that are similar to bread and bagels, whereas extended-brand pairs with low similarity ratings tended to be in categories that are similar to mozzarella cheese and ice cream.

B) Consumers were more likely to purchase both products in extended-brand pairs with ratings comparable to those of bread and bagels than they were to purchase both products in extended-brand pairs with ratings comparable to those of mozzarella cheese and ice cream.

C) The probability that consumers would purchase both products in extended-brand pairs was highest when those products were similar to other products of different brands in the same categories and was lowest when those products were dissimilar to other products of different brands in the same categories.

D) For extended-brand pairs with high similarity ratings, there was a strong positive correlation between the probability that only one product in the pair would be purchased and that particular brand's level of brand recognition.

Dogs Language Detection, v.1

**Find all 3 versions of this question at https://t.me/SATakror*

Neurobiologists Laura Cuaya, Raul Hernández-Pérez, and colleagues investigated the language detection abilities of eighteen dogs raised in similar settings. The researchers monitored the brain activity of Sander (a golden retriever), Alma (a mixed breed), and other dogs while the animals listened to three recordings: one of *The Little Prince* being read in Spanish, the second in Hungarian, and a third made up of short, randomly selected fragments of the first two, scrambled so that they didn't resemble human speech. Each dog was familiar with either Spanish or Hungarian, but not both. The team concluded that the amount of previous language exposure a dog has received may influence its ability to distinguish familiar languages from unfamiliar ones.

33) Which finding from the study, if true, would most directly support the team's conclusion?

A) The difference between the pattern of brain activity a dog showed when hearing the language it was accustomed to and the pattern of brain activity it showed when hearing the language it was not accustomed to was greatest among older dogs.

B) The similarity between the pattern of brain activity a dog showed in response to hearing the scrambled recording and the pattern of brain activity it showed in response to hearing the language it was not accustomed to was greatest among older dogs.

C) Although the dogs' general hearing sensitivity declined with age, dogs of all ages showed more brain activity in response to hearing the language they were accustomed to than in response to hearing the other language.

D) Dogs showed a different pattern of brain activity when hearing the language they were accustomed to than when hearing the scrambled recording, and the difference in brain activity increased with the age of the dog scanned.

Dogs Language Detection, v.2

**Find all 3 versions of this question at https://t.me/SATakror*

Neurobiologists Laura Cuaya, Raul Hernández-Pérez, and colleagues investigated the language detection abilities of eighteen dogs. The researchers monitored the brain activity of Kun-kun (a border collie), Bingo (a mixed breed), and other dogs while the animals listened to three recordings: one of *The Little Prince* being read in Spanish, the second in Hungarian, and a third made up of short, randomly selected fragments of the first two, scrambled so that they didn't resemble human speech. Each dog was familiar with either Spanish or Hungarian, but not both. The team concluded that differences in dogs' anatomical features may affect their ability to distinguish speech from nonspeech.

34) Which finding from the study, if true, would most directly support the team's conclusion?

A) Compared with longer-headed dogs, shorter-headed dogs showed less difference in brain activity when hearing either Spanish or Hungarian than when hearing the scrambled recording.

B) Compared with longer-headed dogs, shorter-headed dogs showed a greater difference in brain activity when hearing the language they were accustomed to than when hearing the other language.

C) The pattern of brain activity that long-headed dogs showed when hearing the scrambled recording was different from the pattern of brain activity that short-headed dogs showed when hearing the language they were accustomed to.

D) Long-headed dogs accustomed to hearing Spanish tended to show more brain activity when hearing Spanish than long-headed dogs accustomed to hearing Hungarian showed when hearing Hungarian.

Dogs Language Detection, v.3

**Find all 3 versions of this question at https://t.me/SATakror*

Neurobiologists Laura Cuaya, Raúl Hernández-Pérez, and colleagues investigated the language detection abilities of eighteen dogs of various ages. The researchers monitored the brain activity of Maverick (a 117-month-old border collie), Mini (a 126-month-old mixed breed), and other dogs while the animals listened to three recordings: one of *The Little Prince* being read in Spanish, the second in Hungarian, and a third made up of short, randomly selected fragments of the first two, scrambled so that they didn't resemble human speech. Each of the dogs was familiar with either Spanish or Hungarian, but not both. The team concluded that the younger the dog, the worse it may be at differentiating between familiar and unfamiliar languages.

35) Which finding from the study, if true, would most directly support the team's conclusion?

A) As the age of the dog scanned decreased, so did the amount of brain activity in response to hearing the language the dog was accustomed to or the other language, but not in response to hearing the scrambled recording.

B) The similarity between the pattern of brain activity a dog showed in response to hearing the scrambled recording and the pattern it showed in response to hearing the language it was not accustomed to was lowest among younger dogs.

C) Dogs showed a different pattern of brain activity when hearing the language they were accustomed to than when hearing the other language, and the difference in brain activity decreased as the age of the dog scanned decreased.

D) Although the dogs' general hearing sensitivity declined with age, dogs of all ages showed more brain activity in response to hearing the language they were accustomed to than in response to hearing the other language.

Invisible Hand, v.1

Eighteenth-century economist Adam Smith is famed for his metaphor of the invisible hand, which he putatively used to illustrate a robust model of how individuals produce aggregate benefits by pursuing their own economic interests. Note "putatively": as Gavin Kennedy has shown, Smith deploys this metaphor only once in his economic writings—to make a narrow point about the then-dominant economic theory of mercantilism—and it was largely ignored until some twentieth-century economists eager to secure an intellectual pedigree for their views elevated it to a fully-fledged paradigm.

36) Which choice best states the main idea of the text?

A) The reputation of Smith's metaphor of the invisible hand is not due to the importance of the metaphor in Smith's work but rather to the promotion of the metaphor by some later economists for their own ends.

B) Smith's metaphor of the invisible hand has been interpreted as a model of how individuals acting in their own interest produce aggregate benefits, but it was intended as a subtle critique of the economic theory of mercantilism.

C) Some twentieth-century economists gave Smith's metaphor of the invisible hand a significance it does not have in Smith's work, but it is nevertheless a useful model of how individuals produce aggregate benefits by pursuing their own economic interests.

D) Although Smith is famed for his metaphor of the invisible hand, the metaphor was largely ignored until economists in the twentieth century came to realize that the metaphor was a robust model that anticipated their own views.

Invisible Hand, v.2

Adam Smith's ideas about "the invisible hand" in economics have been refined and continue to be debated by politicians and economists alike. The basic idea is that the market itself is the best control on the market; for example, prices should not be controlled by the government or any other external agency because a balance between what it costs to produce a good or provide a service and what people are willing to pay for it will produce a fair price, as if guided by an "invisible hand." Thus, one holding to Smith's view of the economy would recommend that the government _____

37) Which choice most logically completes the text?

A) control as much of the economy as possible.

B) appoint appropriate agencies to monitor the economy.

C) permit the economy to find its own balance.

D) force people to accept the costs of goods and services.

Horizontal Gene Transfer, v.1

**Find all 7 versions of this question at https://t.me/SATakror*

Horizontal gene transfer involves the exchange of genetic material between organisms not in a parent-offspring relationship. It is common among prokaryotes (single-celled organisms, such as the bacteria *Enterococcus faecorum* and *Moraxella caviae*). The process can have the effect of increasing bacterial resistance to antibiotics; as such, an understanding of how to prevent horizontal gene transfer might result in the mitigation of dangerous pathogens(organisms that cause disease)

38) Which choice best describes the function of the underlined sentence in the text as a whole?

A) It suggests that a commonly held belief regarding an organism is mistaken.

B) It summarizes the history of research into a biological phenomenon.

C) It illustrates why one organism may pose more of a threat than another.

D) It explains a practical benefit of researching a particular biological process.

Horizontal Gene Transfer, v.2

*Find all 7 versions of this question at https://t.me/SATakror

Horizontal gene transfer involves the exchange of genetic material between organisms not in a parent-offspring relationship. It is common among prokaryotes (single-celled organisms such as the bacteria *Brevicillus borstelensis* and *Massilia timonae*). The process can have the effect of increasing bacterial resistance to antibiotics; as such, an understanding of how to prevent horizontal gene transfer might result in the mitigation of dangerous pathogens (organisms that cause disease).

39) Which choice best states the main idea of the text?

A) A common perception of horizontal gene transfer is inaccurate.

B) Horizontal gene transfer is more common than was initially thought.

C) The results of new research into horizontal gene transfer cast doubt on earlier findings.

D) The study of horizontal gene transfer may yield findings with useful applications.

Horizontal Gene Transfer, v.3

**Find all 7 versions of this question at https://t.me/SATakror*

Horizontal gene transfer involves the exchange of genetic material between organisms not in a parent-offspring relationship. It is common among prokaryotes (single-celled organisms, such as the bacteria *Carnobacterium viridans* and *Lactococcus lactis*). The process can have the effect of increasing bacterial resistance to antibiotics; as such, an understanding of how to prevent horizontal gene transfer might result in the mitigation of dangerous pathogens (organisms that cause disease).

40) Which choice best describes the overall structure of the text?

A) It provides a history of research into a biological process and suggests a direction for further investigation.

B) It explains why an organism poses a threat and then suggests how that threat might be avoided.

C) It depicts two organisms and presents their similarities and differences.

D) It describes a biological process and explains why researching it is important.

Horizontal Gene Transfer, v.4

**Find all 7 versions of this question at https://t.me/SATakror*

Vertical gene transfer involves the transmission of genetic material from a parent to offspring; horizontal gene transfer, on the other hand, involves the exchange of genetic material between organisms not in a parent-offspring relationship. While horizontal gene transfer is common among prokaryotes—single-celled organisms, such as the bacteria *Brevundimonas diminuta* and *Massilia timonensis*—it has rarely been observed among eukaryotes (multicellular organisms). However, new studies suggest that horizontal gene transfer is more common in eukaryotes than originally thought.

41) Which choice best states the function of the underlined sentence in the text as a whole?

A) It contrasts the frequency with which a biological phenomenon has been detected in two categories of organisms.

B) It argues that two biological phenomena are more similar than they may initially appear to be.

C) It explains why a common perception of a biological process is flawed.

D) It indicates a distinction between the mechanics of two kinds of biological processes.

Horizontal Gene Transfer, v.5

**Find all 7 versions of this question at https://t.me/SATakror*

Horizontal gene transfer occurs when an organism of one species acquires genetic material from an organism of another species through nonreproductive means. The genetic material can then be transferred "vertically" in the second species— that is, through reproductive inheritance. Scientist Atma Ivancevic and her team have hypothesized infection by invertebrate parasites as a mechanism of horizontal gene transfer between vertebrate species: while feeding, a parasite could acquire a gene from one host, then relocate to a host from a different vertebrate species and transfer the gene to it in turn.

42) Which choice best describes the function of the underlined portion in the text as a whole?

A) It explains why parasites are less susceptible to horizontal gene transfer than their hosts are.

B) It clarifies why some genes are more likely to be transferred horizontally than others are.

C) It contrasts how horizontal gene transfer occurs among vertebrates with how it occurs among invertebrates.

D) It describes a means by which horizontal gene transfer might occur among vertebrates.

Horizontal Gene Transfer, v.6

**Find all 7 versions of this question at https://t.me/SATakror*

Horizontal gene transfer involves the exchange of genetic material between organisms not in a parent-offspring relationship. It is common among prokaryotes (single-celled organisms, such as the bacteria Enterococcus cecorum and Moraxella caviae). The process can have the effect of increasing bacterial resistance to antibiotics; as such, an understanding of how to prevent horizontal gene transfer might result in the mitigation of dangerous pathogens (organisms that cause disease).

43) Which choice best describes the function of the underlined sentence in the text as a whole?

A) It suggests that a commonly held belief regarding an organism is mistaken.

B) It summarizes the history of research into a biological phenomenon.

C) It illustrates why one organism may pose more of a threat than another

D) It explains a practical benefit of researching a particular biological process.

Horizontal Gene Transfer, v.7

**Find all 7 versions of this question at https://t.me/SATakror*

Vertical gene transfer involves the transmission of genetic material from a parent to offspring: horizontal gene transfer, on the other hand, involves the exchange of genetic material between organisms not in a parent-offspring relationship. While horizontal gene transfer is common among prokaryotes, single-celled organisms such as the bacteria *Carnobacterium viridans* and *Moraxella caviae*, it has rarely been observed among eukaryotes (typically multicellular organisms). However, new studies suggest that horizontal gene transfer is more common in eukaryotes than originally thought.

44) Which choice best states the function of the underlined sentence in the text as a whole?

A) It argues that a particular direction of research concerning horizontal gene transfer is likely to be fruitless.

B) It indicates a distinction between horizontal gene transfer and vertical gene transfer.

C) It implies that a common perception of horizontal gene transfer may be inaccurate.

D) It compares the frequencies with which horizontal gene transfer has been detected in two categories of organisms.

Liberalization, v.1

**Find all 3 versions of this question at https://t.me/SATakror*

In May of 1986, the Philippines liberalized its stock market, meaning that it began allowing foreign individuals and businesses to invest money in Filipino companies. This was part of a wave of stock market liberalizations around the world—Malaysia in 1987, Morocco in 1988, and so on. The standard view among economists at the time was that liberalization would make it easier for companies to raise money from investors. Economist Peter Blair Henry examined the economies of 11 countries that were part of the liberalization wave and found that, on average, companies based in those countries received significant increases in investment in the three years following liberalization, suggesting that _____

45) Which choice most logically completes the text?

A) companies in Malaysia experienced a greater increase in investment following liberalization than did companies in the Philippines.

B) economists who held the standard view of liberalization failed to anticipate some serious negative effects of liberalization.

C) companies did not benefit from liberalization until at least three years after liberalization occurred.

D) empirical evidence was consistent with the scholarly consensus about the consequences of liberalization

Liberalization, v.2

**Find all 3 versions of this question at https://t.me/SATakror*

In November of 1989, Argentina liberalized its stock market, meaning that it began allowing foreign individuals and businesses to invest money in Argentinian companies. This was part of a wave of stock market liberalizations from the mid-1980s through the mid-1990s—Bangladesh in 1991, Nigeria in 1995, and so on. In an analysis of economic data from 1976 to 1993, Ross Levine and Sara Zervos found that liberalization did not lead to enduring increases in investment in companies based in countries that liberalized. Peter Blair Henry, however, found that, on average, investment in companies in liberalized countries increased significantly in the three years following liberalization. Taken together, these results suggest that _____

46) Which choice most logically completes the text?

A) economists' expectations about the effect of liberalization on investment were largely correct.

B) companies typically do not benefit from liberalization until at least three years after liberalization occurs.

C) investment growth is likely to be more consistent in countries that liberalize than in countries that do not.

D) liberalization may provide a boost to investment that fades over time.

Liberalization, v.3

**Find all 3 versions of this question at https://t.me/SATakror*

In June of 1987, South Korean liberalized its stock market, meaning that it began allowing foreign individuals and businesses to invest money in South Korean companies. This was part of a wave of stock markets liberalization from the mid-1980s through the mid-1990s-Brazil in 1988, Venezuela in 1900, and so on. In an analysis of economic data from 1976 to 1993, Ross Levine and Sara Zervos found that liberalization did not lead to enduring increases in investment in companies based in countries that liberalized. Peter Blair Henry, however, found that, on average, investment in companies in liberalized countries increased significantly in the three years following liberation. Taken together, these results suggest that _____

47) Which choice most logically completes the text?

A) it typically takes at least three years for companies to benefit from government policies allowing foreign investment, but governments rarely maintain such policies for that long.

B) companies based in countries that begin allowing foreign investment will probably see short term increases in investment, but that their gains are unlikely to last.

C) expectations of economists about the effect of liberalization on investment were largely correct.

D) investment growth is likely to be more consistent in countries that liberalize than in countries that do not.

Time-oriented expressions, v.1

**Find all 3 versions of this question at https://t.me/SATakror*

The phrase *usa-don sonda* is a future-oriented expression used by members of the Yupno, an Indigenous group in Papua New Guinea, that translates to "next week." Rafael Núñez and colleagues recorded Yupno speakers explaining several of these temporal-related words and phrases and coded each speaker's manual gestures. Previous research has found evidence of Spanish speakers referring to the left/right axis to describe events in time, and many researchers believe that this tendency to make temporal distinctions along imagined linear axes is universal. However, Núñez and colleagues claim that this tendency is not in fact ubiquitous.

48) Which finding, if true, would most directly support Núñez and colleagues' claim?

A) Yupno speakers were observed making temporal gestures both indoors and outdoors, though with greater frequency when indoors.

B) When outdoors and referring to the future, Yupno speakers facing uphill gesture in front of themselves, whereas those facing downhill gesture behind themselves.

C) Yupno speakers typically use their left hand to make temporal gestures regardless of whether the gestures are past oriented or future oriented.

D) Some Yupno grammatical structures used when talking about time are also used in Spanish.

Time-oriented expressions, v.2

**Find all 3 versions of this question at https://t.me/SATakror*

Rafael Núñez and colleagues studied how members of the Yupno, an Indigenous group in Papua New Guinea, conceptualize time. The researchers recorded Yupno speakers explaining certain temporal words and phrases, such as *kalip si ngan*, a past-oriented expression that translates to "a long time ago," and coded each speaker's manual gestures. Previous research has found a tendency in many cultures to make temporal distinctions using spatial concepts and gestures, particularly along egocentric axes (i.e., relative to the orientation of the speaker): for instance, English speakers often refer to the front/back axis to describe events in time. In an anthropology paper, a student claims that the tendency toward ego-based conceptualizations of time is universal.

49) Which finding, if true, would most directly weaken the student's claim?

A) Some Yupno grammatical structures used when talking about time are also used in English.

B) When Yupno speakers who are outdoors use gestures to refer to the future, they point uphill from their current location regardless of which way they are facing.

C) A Yupno speaker points in opposite directions when indicating a past event versus a future event.

D) Although Yupno speakers and English speakers both use gestures to indicate orientation in time, Yupno speakers tend to use fewer gestures overall when speaking than English speakers do.

Time-oriented expressions, v.3

**Find all 3 versions of this question at https://t.me/SATakror*

Rafael Núñez and colleagues studied how members of the Yupno, an indigenous group in Papua New Guinea, conceptualize time in both spoken language and gestures. The researchers recorded Yupno speakers explaining certain temporal words and phrases, such as jare, a past-oriented expression that translates to *"day before yesterday,"* and coded each speaker's manual gestures. Previous research has found a tendency in many cultures to make temporal distinctions along imagined linear axes: for instance, Hebrew speakers often refer to the right/left axis to describe events in time, some researchers believe this tendency is universal, but Núñez and colleagues claim this is not the case.

50) Which finding, if true, would most directly support Núñez and colleagues' claim?

A) Yupno speakers typically use their left hand to make temporal gestures regardless of whether the gestures are past oriented or future oriented.

B) Future-oriented gestures used by Yupno speakers do not, on average, point in the opposite linear direction of past-oriented gestures.

C) Some Yupno grammatical structures used when talking about time are also used in Hebrew.

D) Yupno speakers were observed making temporal gestures both indoors and outdoors, though with greater frequency when indoors.

Mice and Genes, v.1

Students in a biology class investigated why individual house mice (*Mus musculus*) can differ from one another in their susceptibility to cataracts in old age. The students compared wild-type mice and knockout mice, which are mice with specific genes deactivated, when mice of each type were raised in similar naturalistic environments and periodically tested for cataracts. Finding that knockout mice with the gene *Aasdh* deactivated tended to develop cataracts more frequently than did wild-type mice, the students concluded that differences in cataract frequency among house mice in nature are solely attributable to variations in the level of expression of *Aasdh*.

51) Which finding, if true, would most directly weaken the students' conclusion?

A) Some wild-type mice were very similar to the knockout mice with regard to cataract frequency but showed a wide variety of levels of expression of *Aasdh*.

B) The level of expression of *Aasdh* does not appear to affect the functioning of any other genes in house mice.

C) A sampling of house mice captured in natural settings shows that individual mice can differ from one another in the level of expression of *Aasdh*.

D) The mice with *Aasdh* deactivated were identical to the wild-type mice except with regard to cataract frequency.

Mice and Genes, v.2

Students in a biology class investigated why individual house mice (*Mus musculus*) can differ from one another in their circulating iron level. The students compared wild-type mice and knockout mice, which are mice with specific genes deactivated, when mice of each type were placed in similar naturalistic environments and taken for periodic blood sampling. Finding that knockout mice with the gene *Abcg8* deactivated tended to have lower concentrations of iron in their blood than did wild-type mice, the students concluded that differences in circulating iron level among house mice in nature are solely attributable to variations in the level of expression of *Abcg8*.

52) Which finding, if true, would most directly weaken the students' conclusion?

A) A sampling of house mice captured in natural settings shows that individual mice can differ from one another in the level of expression of *Abcg8*.

B) The wild-type mice that were most similar to the knockout mice with regard to circulating iron level also showed the lowest levels of expression of *Abcg8*.

C) The level of expression of *Abcg8* can affect the functioning of other genes that previous studies have suggested may influence circulating iron level.

D) The mice with *Abcg8* deactivated were identical to the wild-type mice except with regard to circulating iron level.

Laughter, v.1

Psychologists Gregory Bryant, Dorsa Amir, and colleagues investigated cross-cultural perceptions of spontaneous (real) laughter and volitional (fake or forced) laughter. Study participants from 21 societies, including those in Japan and South Africa, listened to randomized recordings of 18 spontaneous laughs taken from natural conversations between pairs of women and 18 volitional laughs produced separately by 18 different women in response to an experimenter's instruction to laugh. Analysis of the participants' evaluations of the laughs prompted the team to conclude that the ability to distinguish between spontaneous and volitional laughter appears to be universal across cultures.

53) Which quotation from a psychologist not involved in the team's study would most directly weaken the team's conclusion?

A) "When an individual chooses to produce volitional laughter in a natural social context, the laughter often shares certain acoustic qualities, such as pitch and fluctuation of intensity, with spontaneous laughter."

B) "Although the team considered the average size of communities in each society in the study, that demographic factor was found to have no effect on listeners' identifications of laughter as spontaneous or volitional."

C) "Recent studies in communications have shown that certain acoustic features of spontaneous laughter, such as pitch and intensity, are consistent both within and across societies."

D) "Judgments of spontaneous laughter are often associated with acoustic features such as greater intensity variability and higher pitch."

Laughter, v.2

Psychologists Gregory Bryant, Dorsa Amir, and colleagues investigated cross-cultural perceptions of spontaneous (real) laughter and volitional (fake or forced) laughter. Study participants from 21 societies, including those in Austria and Turkey, listened to randomized recordings of 18 spontaneous laughs taken from natural conversations between pairs of women and 18 volitional laughs produced separately by 18 different women in response to an instruction to laugh. Analysis of the participants' evaluations of the laughs prompted the team to conclude that the ability to distinguish between spontaneous and volitional laughter appears to be universal across cultures.

54) Which potential finding from another study, if true, would most directly strengthen the team's conclusion?

A) Acoustic features of spontaneous laughter, such as pitch and intensity, have measurable variations both within and across societies.

B) There is some evidence of correlation between the average size of communities in a society and judgments of spontaneous and volitional laughter by listeners in that society.

C) Listeners from both Austria and Turkey tend to be able to determine how well two people know each other based on recordings of their conversation.

D) Spontaneous laughter typically has a recognizably higher pitch than volitional laughter, regardless of what society the speakers in a conversation are from.

Hosts and Parasites, v.1

**Find all 3 versions of this question at https://t.me/SATakror*

As complex life cycle parasites, Lecithaster sp. and Capillaria parophrysi require multiple host species throughout their development. Extrapolating from parasite counts on Pacific herring and seven other fish species collected from Puget Sound from 1880 to 2019, Chelsea Wood et al. found that the abundance of three-host parasites, such as Lecithaster sp., negatively correlated with rising average annual sea temperatures; the abundance of two-host parasites, such as C. parophrysi, was largely stable. Noting that fish and other marine vertebrates are especially vulnerable to climate change, Wood et al. observed that all three-host parasites in the study depend on at least two vertebrate species, while all two- host parasites depend on only one, suggesting that _____

55) Which choice most logically completes the text?

A) although Lecithaster sp. and C. parophrysi both require Pacific herring as a host, C. parophrysi was more adversely affected by warming-driven changes in the Pacific herring population than Lecithaster sp. was.

B) population size and density of the eight fish species examined in the study were likely largely unaffected by rising sea temperatures.

C) warming-induced population changes among Puget Sound vertebrates may have suppressed some parasite populations, but there were no changes among the eight studied host species that affected two-host parasite abundance.

D) a parasite's sensitivity to warming temperatures is negatively correlated with the number of unique vertebrate species it depends on throughout its life cycle.

Hosts and Parasites, v.2

**Find all 3 versions of this question at https://t.me/SATakror*

Chelsea Wood et al. tracked temperature-driven changes in the abundance of Gonocerca sp. (a complex life cycle parasite, or CLP, that requires three host species throughout its life cycle), Gyrodactylus sp. (a directly transmitted parasite. which requires only one host species), and 83 other parasite taxa found on eight fish species. CLPs are transmitted when an infected host is ingested by an individual of another species, typically shielding CLPs from the external environment, whereas directly transmitted parasites are exposed to external conditions during transmission. However, Wood et al. found that three host CLP abundance decreased as sea temperatures rose, whereas directly transmitted parasite abundance was largely stable, suggesting that _____

56) Which choice most logically completes the text?

A) as the number of host species involved in a parasite's transmission increases, the parasite is better protected against rising temperatures.

B) directly transmitted parasites identified in the study were more likely to use transmission strategies that shield them from warming temperatures than were three-host CLPs.

C) CLPs primarily transmitted by ingestion were less dependent on host species adversely affected by warming temperatures than were CLPS that use other transmission strategies.

D) any advantages that the transmission strategy used by three-host CLPs may have conferred did not completely offset the negative effects of other temperature-driven factors on CLP abundance.

Hosts and Parasites, v.3

**Find all 3 versions of this question at https://t.me/SATakror*

To measure changes in parasite abundance over time, Chelsea Wood and colleagues counted parasite individuals preserved on specimens of striped sea perch, surf smelt, and six other fish species collected from Puget Sound between 1880 and 2019. Using statistical models to estimate historical populations, the researchers determined that for every 1°C increase in annual average sea surface temperature, the abundance of complex life cycle parasites, such as Lecithaster sp. that require at least three unique host species throughout their life cycle, decreased by 38%. However, the abundance of Bomolochus bellones and other directly transmitted parasites, which require only one host species, was essentially unchanged. These findings suggest that _____

57) Which choice most logically completes the text?

A) Lecithaster sp. abundance decreased by 38% over the period studied, whereas B. bellones abundance did not.

B) parasites that rely exclusively on either striped sea perch or surf smelt are more sensitive to rising temperatures than are parasites that can infect both species throughout their life cycles.

C) dependency on only a single host species may confer on parasites some resilience to rising sea surface temperatures.

D) as the number of hosts that complex life cycle parasites require increases, the parasites' tolerance for rising sea surface temperatures decreases proportionally.

Historical DNA, v.1

**Find all 5 versions of this question at https://t.me/SATakror*

Reyjane P. Oliveira and colleagues relied on historical DNA (hDNA)—genomic data incidentally preserved in specimens housed in natural history collections—to investigate the evolutionary trajectory of the Malagasy grass genus Chasechloa. Although this approach offers unique benefits, such as the ability to ascertain the biogeographical origins of organisms, it remains a relatively underutilized resource because hDNA is often to some extent degraded, a situation not easily remediable under current methodological paradigms and with extant DNA extraction and analysis technologies.

58) Information in the text best supports which statement about hDNA?

A) It may yield insights that other types of genomic data cannot.

B) It may be underused because of its controversial status among scientists.

C) It has thus far proved valuable mainly to researchers studying flora.

D) It tends to be much more degraded than other types of DNA of comparable age.

Historical DNA, v.2

**Find all 5 versions of this question at https://t.me/SATakror*

Allison Q. Byrne and colleagues relied on historical DNA (hDNA)—genomic data incidentally preserved in specimens housed in collections such as those at the Smithsonian National Museum of Natural History—to investigate the evolutionary origins of a pathogen affecting amphibians. Although this approach can yield many insights about the biological past, it remains a relatively underutilized resource in part because DNA is often to some extent degraded, a situation not easily remediable under current methodological paradigms and with extant DNA extraction and analysis technologies.

59) Which choice best describes the overall structure of the text?

A) It evaluates a research methodology used in a particular study, then explains how scientists may overcome difficulties inherent to that methodology.

B) It summarizes the findings of a scientific study, then discusses the possibility of other researchers replicating those findings in future studies.

C) It exemplifies a common method of genomic data analysis, then details the difficulties in adapting that method to new circumstances.

D) It presents a scientific study that relied on a particular approach, then describes a barrier to the widespread adoption of that approach.

Historical DNA, v.3

**Find all 5 versions of this question at https://t.me/SATakror*

60

Svante Pääbo and other researchers studying the history of organisms have long utilized ancient hDNA—DNA recovered from ancient organic material that has been preserved under natural conditions. However, Reyjane P. Oliveira and colleagues' 2017 study of the evolutionary trajectory of the Malagasy grass genus Chasechloa instead relied on historical DNA—genomic data incidentally preserved in specimens that are housed in natural history collections—thus capitalizing on the research potential offered by a vast but hitherto relatively underutilized source of insight into the biological past.

60) Which choice best describes the function of the underlined portion in the text as a whole?

A) It specifies potential applications of the approach that Oliveira and colleagues used in their study.

B) It offers commentary on the significance of the approach that Oliveira and colleagues used for their study.

C) It emphasizes the importance of Oliveira and colleagues findings about the DNA of flora.

D) It explains why the research methodology selected by Oliveira and colleagues is not widely used.

Historical DNA, v.4

**Find all 5 versions of this question at https://t.me/SATakror*

Nicolas Dussex and colleagues relied on historical DNA (hDNA)—genomic data incidentally preserved in specimens housed in natural history collections—to investigate the evolutionary trajectory of the kakapo parrot (Strigops habroptilus). Although this approach offers unique benefits, such as the ability to sample genomic material from extinct species, it remains relatively underutilized because archival specimens are sometimes stored in ways that compromise DNA quality, a situation not easily remediable under current methodological paradigms and with extant DNA extraction and analysis technologies.

61) What does the text most strongly suggest about specimens from natural history collections?

A) While they were used in a study that made an important scientific discovery, they are generally of marginal value as sources of genomic data.

B) Because of their often-deteriorated condition, they tend to yield genomic data that are time-consuming to extract and interpret.

C) They are primarily used as sources of genomic data by scientists studying organisms that have undergone major evolutionary change.

D) While they may contain valuable genomic data, not all of them can yield usable hDNA.

Historical DNA, v.5

**Find all 5 versions of this question at https://t.me/SATakror*

Kirsten Bos and other researchers studying the history of organisms have long utilized ancient DNA — DNA recovered from ancient organic material that has been preserved under natural conditions. However, Maria C. Avila-Arcos and colleagues' 2013 study of the evolutionary trajectory of the koala retrovirus instead relied on historical DNA (hDNA) — genomic data incidentally preserved in specimens that are housed in natural history collections — thus capitalizing on the research potential offered by a vast but hitherto relatively underutilized source of insight into the biological past.

62) Which choice best describes the overall structure of the text?

A) It provides an overview of a technique traditionally used by scientists, then asserts that an alternative technique is superior to it.

B) It compares two distinct methods of analyzing genomic data, then explains why the second method is relatively underused.

C) It describes two different approaches used by researchers, then emphasizes how promising one of those approaches is.

D) It summarizes previous research on a topic, then illustrates how a group of scientists used a new method to challenge findings produced by that research.

Nautilids, v.1

**Find all 3 versions of this question at https://t.me/SATakror*

Over time, marine mollusks called nautilids continually add segments to their shells, and the chemical composition of each segment is influenced by the available materials in the surrounding water when the segment forms. Paleontologist Amane Tajika and colleagues analyzed the septa (the walls between shell chambers) of one male (samples labeled "M") and one female (samples labeled "F") nautilid, finding that sample F07 had the same concentration of oxygen-18 as did water approximately 135 meters deep and that the oxygen-18 concentration in sample M28 matched that of water at approximately 365 meters deep. Tajika et al., concluded that sample F07 formed at a higher point in the water column than did M28.

63

63) Which finding, if true, would most directly support the researchers' conclusion?

A) In nautilids, sex predicts septa growth rate more strongly than does oxygen-18 concentration in the surrounding water.

B) Among nautilids, there is no effect of sex on the uptake of oxygen-18 from water or its deposition in septa.

C) Male nautilids tend to show higher concentrations of oxygen-18 than do female nautilids inhabiting the same point in the water column.

D) Regardless of sex, nautilids spend most of their lives at depths of 50 meters or greater, where oxygen-18 concentrations typically differ from those higher in the water column.

Nautilids, v.2

**Find all 3 versions of this question at https://t.me/SATakror*

Nautilids are marine mollusks that begin growing their shells before emerging from their eggs and continue to add shell segments throughout their lifetimes. The walls between their chambers are called septa, and the deeper the water in which a septum forms, the greater the concentration of the isotope oxygen-18 the septum will contain. Since temperature falls as depth increases, if the area of ocean a nautilid inhabits is known, this isotopic signature can reveal the temperature at which its septa formed. Paleontologist Amane Tajika and colleagues examined each of the septa in two nautilid shells and concluded that septum sample M23 formed at a temperature of 14.0°C whereas sample F02 formed at 20.8°C.

64) Which finding, if true, would most directly weaken the researchers' conclusion?

A) At a depth of 140 meters in the nautilids' habitat, the water is 20.8°C and the concentration of oxygen-18 is equal to that in sample F02.

B) The concentration of oxygen-18 in F02 is higher than that in M23

C) The concentration of oxygen-18 in F02 is lower than that in M23.

D) At a depth of 385 meters in the nautilids' habitat, the water is 14.0°C and the concentration of oxygen-18 is equal to that in sample M23.

Nautilids, v.3

**Find all 3 versions of this question at https://t.me/SATakror*

Paleontologist Amane Tajika and colleagues analyzed the shells of two marine mollusks called nautilids that were collected from waters near New Caledonia. Sample F13 and other shell sections formed during adulthood suggest that mature New Caledonian nautilids live at the same depths as mature Fijian nautilids. However, sample Mo6, which formed while the nautilid was still in its egg, had a chemical signature suggesting the egg had been laid in water 120 meters deep, while Fijian nautilids typically lay eggs at depths of at least 170 meters. Because water temperature decreases as depth increases, a biology student hypothesized that the New Caledonian nautilids lay their eggs in warmer water than Fijian nautilids do.

65) Which finding, if true, would most directly weaken the student's hypothesis?

A) New Caledonia nautilids tend to migrate upward to warmer water immediately after hatching, whereas Fijian nautilids do not.

B) The water temperature at a given depth tends to be higher in waters near Fijian it is at that depth in waters near New Caledonia.

C) There aren't enough suitable hatching areas at a depth of at least 170 meters in New Caledonia for nautilids to lay their eggs at that depth.

D) The water temperature at a given depth tends to be lower in waters near Fiji than it is at that depth in waters near New Caledonia.

The Seine, v.1

The following text is adapted from Guy de Maupassant's 1884 short story "A Recollection," from the collection Guy de Maupassant Short Stories (translated by Albert M.C. McMaster et al. in 1903). The narrator is taking a boat down the Seine river from Paris, France, to the surrounding countryside.

I took up a position in the bows [front of the boat], standing up and looking at the quays, the trees, the houses, and the bridges disappearing behind us. And suddenly I perceived the great viaduct of Point du Jour which blocked the river. It was the end of Paris, the beginning of the country, and behind the double row of arches the Seine, suddenly spreading out as though it had regained space and liberty, became all at once the peaceful river which flows through the plains, alongside the wooded hills, amid the meadows, along the edge of the forests.

66) Based on the text, which choice best expresses the narrator's characterization of the Seine?

A) Though the Seine is calm during most of the journey, the arches of the Point du Jour created a small area of heavy current.

B) The Seine is beautiful but not as beautiful as Paris.

C) As the journey progressed from Paris to the countryside, the waters of the Seine gradually cleared.

D) The waters of the Seine are more confined and rough in Paris than they are in the countryside.

The Seine, v.2

The following text is adapted from Guy de Maupassant's 1884 short story "A Recollection," from the collection Guy de Maupassant Short Stories (translated by Albert M.C. McMaster et al. in 1903). The narrator is taking a boat down the Seine river from Paris, France, to the surrounding countryside.

I took up a position in the bows [front of the boat], standing up and looking at the quays, the trees, the houses, and the bridges disappearing behind us. And suddenly I perceived the great viaduct of Point du Jour which blocked the river. It was the end of Paris, the beginning of the country, and behind the double row of arches the Seine, suddenly spreading out as though it had regained space and liberty, became all at once the peaceful river which flows through the plains, alongside the wooded hills, amid the meadows, along the edge of the forests.

67) Which choice best states the main idea of the text?

A) The narrator contrasts his feelings of apprehension with the apparent tranquility of the river as the boat enters the countryside.

B) The narrator is anxious for the trip to the countryside to be over as quickly as possible, and is frustrated that it is taking so long.

C) Though the narrator did not expect to be leaving the city on this trip, he is nonetheless excited to undertake this new adventure.

D) The narrator perceives a parallel between the shift from the urban environment to the countryside and the changes in the features of the river itself.

Representation of Styles, v.1

Text 1
Scholarship today overrepresents experimentally fragmented narrative structures, such as that of William Faulkner's *As I Lay Dying*, beyond the degree to which they actually influenced fiction in the United States during the modernist period (roughly 1900-1945). Meanwhile, Ellen Glasgow's *Barren Ground*, whose coherent, linear narrative structure recalls the fiction of previous centuries, attracts woefully little attention from scholars of modernism.

Text 2
Distant reading, or computer-assisted quantitative analysis of massive collections of digitized texts, can reveal stylistic elements that have heretofore escaped notice, despite being shared by numerous texts from the modernist period. For too long, scholars have focused on narrative fragmentation versus coherence, inhibiting inquiry into other points of stylistic correspondence among works that would enrich our understanding of the modernist canon.

68) Based on the texts, both the author of Text 1 and the author of Text 2 would most likely agree with which statement about scholarship on the modernist period in the United States?

A) Its primary methods for analyzing fiction written in the period are growing obsolete as computer technology advances.

B) Instead of engaging in unproductive debates, it should work to rehabilitate the reputations of neglected modernist works.

C) At present, it only partially captures the stylistic dimensions of the fiction written in the period.

D) It must widen its focus to include aspects of modernist fiction beyond style, a productive but overrepresented site of inquiry.

Representation of Styles, v.2

Text 1
Scholarship today overrepresents formal experimentation, such as William Carlos Williams' use of minimalistic, image-based structures, well beyond the degree to which it actually influenced US poetry during the modernist period (roughly 1900-1945). Meanwhile, the work of Countee Cullen, who relied on conventional poetic forms associated with previous literary periods, attracts woefully little attention from scholars of modernism.

Text 2
Distant reading, or computer-assisted quantitative analysis of massive collections of digitized texts, can reveal stylistic elements that have heretofore escaped notice, despite being shared by numerous texts from the modernist period. For too long, scholars have focused on experimental versus conventional poetic forms, inhibiting inquiry into other points of stylistic correspondence among poems that would enrich our understanding of the modernist canon.

69) Based on the texts, how would the author of Text 2 most likely regard the emphasis that Text 1 places on poetic form during the modernist period?

A) As having the effect of forestalling consideration of the full stylistic dimensions of poetry written in the period

B) As overestimating the impact that poetic form, whether experimental or conventional, has on the reading experience

C) As being founded on a misconception about modernist poetry that has been called into question by distant reading

D) As complementing the study of other stylistic features that distinguish the modernist period

Convergent Evolution, v.1

**Find all 5 versions of this question at https://t.me/SATakror*

Meredith F. Protas and colleagues have explored how convergent evolution – a phenomenon that occurs when the same trait evolves independently in two reproductively separate lineages – can result from a genetic mechanism shared by both lineages. Meanwhile, Cynthia C. Steiner and colleagues have investigated how convergence occurs through different genetic mechanisms, but the relative prevalence of convergence through shared and different genetic processes is still poorly understood. This motivated biologists Delbert A. Green II and Cassandra G. Extavour to evaluate both types of convergence in a single study for their 2012 paper.

70) Which choice best states the main idea of the text?

A) Green and Extavour's study was conducted using data from the studies by Protas and colleagues and Steiner and colleagues.

B) Green and Extavour's study addresses convergent evolution more comprehensively than the studies by Protas and colleagues and Steiner and colleagues do.

C) Both the study by Protas and colleagues and that by Green and Extavour compare convergence through shared genetic mechanisms to convergence through different genetic mechanisms.

D) Before Green and Extavour's study, convergent evolution was assumed to require a shared genetic mechanism between two lineages that share the trait.

Convergent Evolution, v.2

**Find all 5 versions of this question at https://t.me/SATakror*

Elio Sucena and colleagues have explored how convergent evolution—a phenomenon that occurs when the same trait evolves independently in two reproductively separate lineages—can result from a genetic mechanism shared by both lineages. Meanwhile, Michael D. Shapiro and colleagues have investigated how convergence occurs through different genetic mechanisms, but the relative prevalence of convergence through shared and different genetic processes is still poorly understood. This motivated biologists Delbert A. Green II and Cassandra G. Extavour to evaluate both types of convergence in a single study for their 2012 paper.

71) Which choice best states the main purpose of the text?

A) To suggest that Green and Extavour's conclusions would be more persuasive if they incorporated findings from the studies by Sucena and colleagues and Shapiro and colleagues.

B) To argue that Green and Extavour's study revealed a previously unrecognized issue that evolutionary biologists need to address.

C) To summarize Green and Extavour's criticism of the study by Sucena and colleagues.

D) To situate Green and Extavour's study in the context of their field.

Convergent Evolution, v.3

**Find all 5 versions of this question at https://t.me/SATakror*

Elio Sucena and colleagues have explored how convergent evolution—a phenomenon that occurs when the same trait evolves independently in two reproductively separate lineages—can result from a genetic mechanism shared by both lineages. Meanwhile, Patricia J. Wittkopp and colleagues have investigated how convergence occurs through different genetic mechanisms, but the relative prevalence of convergence through shared and different genetic processes is still poorly understood. This motivated biologists Delbert A. Green II and Cassandra G. Extavour to evaluate both types of convergence in a single study for their 2012 paper.

72) Which choice best states the overall structure of the text?

A) It explains a widespread assumption about a phenomenon, describes two studies that are based on that assumption, and explains why this assumption is unfounded.

B) It discusses a particular study of a phenomenon, elaborates on how other studies have approached this phenomenon in the past, and critiques the first study's approach.

C) It details the development of a field of scientific inquiry, gives examples of three studies that were important to this development, and speculates on how future studies can develop these ideas.

D) It describes a phenomenon, provides examples of two studies of that phenomenon, and mentions a third study of that phenomenon.

Convergent Evolution, v.4

**Find all 5 versions of this question at https://t.me/SATakror*

Benjamin Prud'homme and colleagues have explored how convergent evolution—a phenomenon that occurs when the same trait evolves independently in two reproductively separate lineages—can result from a genetic mechanism shared by both lineages. Meanwhile, Cynthia C. Steiner and colleagues have investigated how convergence occurs through different genetic mechanisms. However, the relative prevalence of convergence through shared and different genetic processes is still poorly understood. This motivated biologists Delbert A. Green II and Cassandra G. Extavour to evaluate both types of convergence in a single study for their 2012 paper.

73) Which choice best states the function of the underlined sentence in the text as a whole?

A) It introduces researchers who will be discussed in greater detail later in the text.

B) It gives an example of how some scientists had studied a phenomenon before another study mentioned later in the text was conducted.

C) It outlines a study that was influenced by the researchers mentioned later in the text.

D) It clarifies a concept that is unclear in some of the studies mentioned in the text.

Convergent Evolution, v.5

**Find all 5 versions of this question at https://t.me/SATakror*

Benjamin Prud'homme and colleagues have explored how convergent evolution—a phenomenon that occurs when the same trait evolves independently in two reproductively separate lineages—can result from a genetic mechanism shared by both lineages. Meanwhile, Patricia J. Wittkopp and colleagues have investigated how convergence occurs through different genetic mechanisms, but the relative prevalence of convergence through shared and different genetic processes is still poorly understood. This motivated biologists Delbert A. Green II and Cassandra G. Extavour to evaluate both types of convergence in a single study for their 2012 paper.

74) Which choice best states the function of the underlined portion in the text as a whole?

A) It provides examples of how a phenomenon was studied by scientists in the field before Green and Extavour's study.

B) It gives a basic description of a phenomenon that is central to the discussion that follows.

C) It clarifies a concept that the author implies was unclear in the studies mentioned in the text.

D) It introduces a method of scientific analysis that is discussed in greater detail later in the text.

Plants vs. Respiration, v.1

Periods of subfreezing temperatures in Alaska have been growing shorter in response to climate warming, potentially enabling increased carbon dioxide (CO2) absorption through greater productivity of creeping dogwood (Cornus canadensis) plants and other vegetation, but also potentially enabling Early snow melt reduces the amount of soil moisture available for the increased CO2 output through greater heterotrophic respiration (CO2 generated by the activity of soil microorganisms). Hydrologist Yonghong Yi and her colleagues developed a model incorporating numerous inputs – years of solar radiation and snow cover data among them – to evaluate the effects of warming on the CO2 balance in Alaska, concluding that net CO2 is likely to increase if warming hastens spring snow melt.

75) Which finding, if true, would most directly support the researchers' conclusion?

A) Early snow melt reduces the amount of soil moisture available for the growth of plant species such as C. canadensis, and it raises the rate of heterotrophic respiration.

B) Early snow melt extends the period in which plant species such as C. canadensis can absorb CO2, but has no effect on the rate of heterotrophic respiration.

C) Early snow melt slows the growth of plant species such as C. canadensis by reducing soil insulation, and it suppresses heterotrophic respiration.

D) Early snow melt amplifies the effect of solar radiation on the growth of plant species such as C. canadensis and on the rate of heterotrophic respiration.

Plants vs. Respiration, v.2

**Find all 7 versions of this question at https://t.me/SATakror*

Soil thawing in Alaska has been accelerating as a result of climate changes, potentially enabling increased carbon dioxide (CO2) absorption through greater productivity of arctic bramble (Rubus arcticus) plants and other vegetation, but also potentially enabling increased CO2 output through greater heterotrophic respiration (CO2 generated by the activity of soil microorganisms). Hydrologist Yonghong Yi and her colleagues developed a model incorporating numerous inputs—years of soil temperature and snow cover data among them—to evaluate the effects of climate changes on the CO2 balance in Alaska, concluding that net CO2 is likely to increase if seasonal snow cover arrives earlier relative to the onset of soil surface freezing.

76) Which finding, if true, would most directly support the researchers' conclusion?

A) The effect of soil temperature on the growth of vegetation and on the rate of heterotrophic respiration is stronger during the period of seasonal snow cover than during the period without snow cover.

B) Relatively early snow cover reduces the amount of soil moisture available for the growth of plant species such as R. arcticus and lowers the rate of heterotrophic respiration.

C) Seasonal snow cover tends to persist longer in areas of relatively low vegetation growth and high heterotrophic respiration than in areas of relatively high vegetation growth and low heterotrophic respiration.

D) The soil insulation provided by snow cover enables heterotrophic respiration to continue during a period in which plant species such as R. arcticus are typically not growing.

Traceability vs. Consistency, v.1

Callie W. Babbitt, Hema Madaka, and colleagues assembled a database of materials used in consumer electronics by studying products in the lab and by gathering data from similar product studies. The team gave each of these studies a rating for level of traceability (with a higher rating for clearer descriptions of procedures) and for category consistency (with a higher rating for using materials categories more closely aligned with the categories in the team's database). Based on these ratings, a second research team concluded that the methodology was better explained in a study by Greg L. Kozak and Gregory A. Keoleian than it was in a study by Martin Streicher-Porte and colleagues.

77) Which finding, if true, would most directly challenge the second research team's conclusion?

A) The study by Kozak and Keoleian had a lower consistency rating than the study by Streicher-Porte and colleagues did.

B) The study by Kozak and Keoleian had a high consistency rating and a high traceability rating.

C) The study by Streicher-Porte and colleagues had a medium consistency rating and a medium traceability rating.

D) The study by Kozak and Keoleian had a lower traceability rating than the study by Streicher-Porte and colleagues did.

Traceability vs. Consistency, v.2

Callie W. Babbitt, Hema Madaka, and colleagues assembled a database of materials used in consumer electronics by studying products in the lab and by gathering data from other similar product studies. The team rated each study on its level of detail (with a higher rating for reported data with more detail) and its category consistency (with a higher rating for using materials categories more closely aligned with the categories in the team's database). Based on these ratings, a second research team concluded that the data in a study by Maria Leet Socolof and colleagues would need less adjustment to fit the classifications in the database than the data in a study by Martin Streicher-Porte and colleagues did.

78) Which finding, if true, would most directly support the second research team's conclusion?

A) The study by Socolof and colleagues had a higher consistency rating than the study by Streicher-Porte and colleagues did.

B) The study by Socolof and colleagues had a high consistency rating and a medium detail rating.

C) The study by Socolof and colleagues had a higher detail rating than the study by Streicher-Porte and colleagues did.

D) The study by Streicher-Porte and colleagues had a medium consistency rating and a medium detail rating.

Environment and Consumers, v.1

Saeed M.Z.A. Tarabieh conducted a study of consumer attitudes toward Jordanian food and beverage companies and found that for consumers who value environmental conservation, their likelihood of purchasing a product decreased when their perception of the product's risk of causing environmental harm increased. Subsequently, other researchers conducted a study of various demographic groups in Greece, investigating participants' intentions to purchase a new piece of furniture, and found that, on average, suburban residents had the highest perception among all the demographic groups in the study of the environmental risks of the piece of furniture. Assuming that the results of Tarabieh's study are broadly applicable, this finding suggests that _____

79) Which choice most logically completes the text?

A) The new piece of furniture is more appealing to suburban residents than other similar products on the market are.

B) Suburban residents might be less likely than participants in the other demographic groups to purchase the piece of furniture.

C) There is not a meaningful difference in the average likelihood of purchasing environmentally friendly products among the demographic groups included in the study.

D) Suburban residents likely prioritize other factors over a product's environmental sustainability when making purchasing decisions.

Environment and Consumers, v.2

In a 2018 study, Deepak Jaiswal and Rishi Kant found that consumers' knowledge of environmental issues had no effect on the likelihood that the consumers would purchase environmentally friendly products. Since this study was based on fewer than 400 young adults in India, however, doubts have been raised about how reliable and representative the findings are. To better understand the issue, Wencan Zhuang and colleagues analyzed the results of 54 studies of eco-friendly consumer behavior, such as a 2018 study from Indonesia that included 916 participants and a 2018 study from India with 202 participants. Taking all 54 studies together, Zhuang and colleagues found a significant positive effect of environmental knowledge on eco-friendly purchasing decisions, suggesting that _____

80) Which choice most logically completes the text?

A) A sample size of 202 may be sufficient to make reliable conclusions about the relationship between knowledge of environmental issues and purchasing decisions.

B) Concerns about the broad applicability of Jaiswal and Kant's conclusion were justified.

C) The number of participants in Jaiswal and Kant's study was far below the number of participants in most studies of purchasing decisions.

D) Jaiswal and Kant's methodology was more precise than the methodology used in the 2018 study from Indonesia.

Neighborhoods and Languages, v.1

**Find all 3 versions of this question at https://t.me/SATakror*

Over 600 languages are spoken in New York City in addition to English – one can find Acehnese spoken in the neighborhood of Astoria, for example, or Hindi in Gramercy Park. English is the most common, with 65% of New Yorkers speaking it at home; Spanish is second, at 20%. A linguist hypothesizes that a Spanish dialect – a variation different from the parent language in some of its vocabulary, pronunciation, and grammar – has developed in New York City, partly due to its linguistic interaction with English. Previous research shows that younger speakers of non-majority languages are more likely to borrow words from the majority language of a region than older speakers. If the linguist's hypothesis is correct, it is therefore likely to be the case that _____

81) Which choice most logically completes the text?

A) Younger speakers of Spanish in New York City would be more likely to use a dialect specific to New York City than older speakers living in the same neighborhood would be.

B) The proportion of younger speakers who use a Spanish dialect specific to New York City is higher in neighborhoods where many languages are spoken than in neighborhoods where Spanish and English predominate.

C) Both the number of languages spoken in New York City and the number of dialects of each of those languages will increase over time.

D) Spanish used by younger speakers in New York City tends to have more words borrowed from other non-majority languages than borrowed from English.

Neighborhoods and Languages, v.2

**Find all 3 versions of this question at https://t.me/SATakror*

Over 600 languages are spoken in New York City in addition to English – one can find Bengali spoken in the neighborhood of Parkchester, or Fujianese in Sunset Park. Most speakers of Chinese languages reside in the neighborhood of Flushing (part of New York City's borough of Queens) and in Chinatown, in the borough of Manhattan. New immigrants from north China, where Mandarin is the primary first language, tend to settle in Queens, while new immigrants from south China, where many people speak Cantonese or Fuzhounese as a first language, tend to settle in Manhattan. It can therefore be inferred that _____

82) Which choice most logically completes the text?

A) Languages tend to change more rapidly in areas where many languages are spoken than in areas where few languages are spoken.

B) There is a positive correlation between the physical size of a country and the number of languages spoken in that country.

C) Correlations in a country between languages and regions where they are spoken can replicate themselves in a new country to which the original country's citizens emigrate.

D) Languages spoken by immigrant peoples can differ significantly in vocabulary and pronunciation from those same languages in their country of origin.

Neighborhoods and Languages, v.3

**Find all 3 versions of this question at https://t.me/SATakror*

Over 600 languages are spoken in New York City in addition to English – one can find Aromanian spoken in the neighborhood of Ridgewood, or Hindi in Gramercy Park. Many immigrants from north China, where Mandarin is the primary language, reside in the neighborhood of Flushing (part of New York City's borough of Queens) while those who emigrated from south China, where many people speak Cantonese or Fuzhounese as a first language, tend to reside in Chinatown, in the borough of Manhattan. Among speakers of Chinese languages, those in Flushing tend to speak Mandarin as their primary language while those in Chinatown tend to speak Cantonese or Fuzhounese as their primary language. This is most likely because _____

83) Which choice most logically completes the text?

A) Chinese immigrants who emigrated to New York City many years ago are more likely to speak several Chinese languages than more recent Chinese immigrants.

B) Taken together, there are more Cantonese and Fuzhounese speakers among Chinese immigrants in New York City than there are Mandarin speakers.

C) Chinese immigrants regularly change their residences between Queens and Manhattan after they emigrate, rather than staying in one borough.

D) There are geographic differences in primary language within China that are reflected in the settlement patterns of Chinese immigrants in New York City.

In The Name of Salome, v.1

The following text is from Julia Alvarez's 2000 novel *In the Name of Salomé*. Salomé, a poet, is hosting guests in the front parlor of her family home, and Ramona is her sister. A salon is a social gathering for the exploration of intellectual ideas.

It was evening when the two men got up to leave. Tía Ana had already come into the room several times to see if these guests had departed yet. The front parlor had always been her special province, as she used it for her little infirmary school. Now, every evening, it turned into Salomé's salon, as Ramona called it, and it was never in order for its transformation back to a classroom the following morning.

84) Based on the text, what most likely motivates Tía Ana's behavior during Salomé's salon?

A) She is frustrated because she needs assistance elsewhere in the house, but Salomé is unavailable while entertaining the guests.

B) She is anxious for the gathering to disperse so that she can ready the space for her own needs.

C) She is impatient to share her plans to start a new school with the guests and hopes they will support her.

D) She finds the guests to be dull and is eager to find a way to leave the salon.

In The Name of Salome, v.2

The following text is from Julia Alvarez's 2000 novel *In the Name of Salomé*. Salomé, a poet, is hosting guests in the front parlor of her family home, and Ramona is her sister. A salon is a social gathering for the exploration of intellectual ideas.

It was evening when the two men got up to leave. Tía Ana had already come into the room several times to see if these guests had departed yet. The front parlor had always been her special province, as she used it for her little infirmary school. Now, every evening, it turned into Salomé's salon, as Ramona called it, and it was never in order for its transformation back to a classroom the following morning.

85) Which choice best describes the function of the reference to the parlor as Tía Ana's "special province"?

A) It reveals that Tía Ana feels as if Salomé has betrayed her trust by allowing guests into space she considers her own.

B) It indicates the joy that Tía Ana feels when she uses her classroom for varying purposes.

C) It characterizes the room as a place where Tía Ana can go to escape social pressures.

D) It introduces an idea that helps explain Tía Ana's apparent eagerness for Salomé's guests to leave the space.

Fish and Low-visibility, v.1

**Find all 7 versions of this question at https://t.me/SATakror*

In the aquatic kingdom, three species of fish exhibit distinct head and mouth morphology: the Broadface Chromis latus, the Mediumface Chromis mediocris, and the Narrowface Chromis angustus. Historically, it has been observed that under low visibility conditions in water, the Narrowface Chromis exhibits the highest sensitivity to changes in water pressure, indicating a high ability to sense waves and pressure changes. It is followed by the Mediumface Chromis and then the Broadface Chromis. A scientist hypothesized that in vortex conditions where rapid water movement significantly affects all fish's visual capabilities, fish with a stronger ability to sense water waves and pressure can better evade obstacles in the water than other fish, effortlessly avoiding obstructions. To test this hypothesis, researchers placed the three species into a vortex environment, introduced a series of obstacles, and recorded the success rate at which each species avoided the obstacles.

86) Which of the following experimental findings, if true, would most directly challenge the scientist's hypothesis?

A) The Chromis latus, with the widest head and mouth, successfully evades half of the obstacles.

B) Some Narrowface Chromis angustus exhibited a lower success rate at avoiding obstacles than other Chromis angustus.

C) Despite their low sensitivity to water pressure, some Broadface Chromis latus successfully circumvented all of the obstacles.

D) The Mediumface Chromis mediocris and the Broadface Chromis latus showed similar rates of collisions with obstacles, both struggling to avoid obstructions.

Fish and Low-visibility, v.2

**Find all 4 versions of this question at https://t.me/SATakror*

Water flowing around an obstruction creates vortices (patterns of swirls) of varying size; by detecting the vortices, fish can determine the size and position of the obstruction. Testing by Yuzo R. Yanagisura, Otar Akanyeti, and James C. Liao using models of three head shapes—narrow (low ratio of width to length), intermediate, and wide (high ratio of width to length)—showed that for medium-sized vortices, fish with wide heads would be least able to distinguish between vortices and general turbulence in the water. A second research team has therefore hypothesized that in low-visibility conditions, wider-headed fish will be less likely than narrower-headed fish to detect obstructions.

87) Which finding, if true, would most directly support the second research team's hypothesis?

A) A study using obstructions that created medium-sized vortices in low-visibility conditions found that some specimens of dusky smooth-hound (Mustelus canis), which has a relatively narrow head, bumped into the obstructions more often than other specimens of the same fish did.

B) A study using obstructions that created medium-sized vortices in low-visibility conditions found that the wider-headed bristlemouth (Chaetostoma yurbiense) bumped into obstructions more often than the narrower-headed dusky smooth-hound (Mustelus canis) did.

C) A study using obstructions that created medium-sized vortices in low-visibility conditions found that the narrower-headed dusky smooth-hound (Mustelus canis) bumped into the obstructions just as often as the wider-headed bristlemouth (Chaetostoma yurbiense) did.

D) A study using obstructions that created medium-sized vortices in low-visibility conditions found that the bristlemouth (Chaetostoma yurbinese), which has a relatively wide head, bumped into more than half of the obstructions.

Fish and Low-visibility, v.3

**Find all 7 versions of this question at https://t.me/SATakror*

88

To detect information about water flow, fish have sensors running from the snout tip down the sides of the head. Yuzo R. Yanagisuru, Otar Akanyeti, and James C. Liao conducted tests to find where the difference in pressure is greatest between two adjacent sensors because, according to the researchers, as these pressure differences increase, so does the amount of information available to the fish. Using the ratio of head width to length, they found that the greatest pressure difference is closer to the snout for narrower heads (lower ratio of width to length) and farther from the snout for wider heads (higher ratio of width to length). Based on this finding, a second team of researchers has hypothesized that the sensors where information is greatest are likely more sensitive than the rest.

88) Which finding, if true, would most directly support the second research team's hypothesis?

A) The longnose gar (Lepisosteus osseus) has a much narrower head than the devil catfish (Bagarius bagarius), and the most sensitive sensors of the longnose gar are closer to the snout than are those of the devil catfish.

B) The longnose gar (Lepisosteus osseus) has a much narrower head than the devil catfish (Bagarius bagarius), and the most sensitive sensors for both are very close to their snouts.

C) The longnose gar (Lepisosteus osseus) has a much narrower head than the devil catfish (Bagarius bagarius), and while the sensors nearest the snout for the longnose gar are more sensitive than the others, for the devil catfish all sensors are equally sensitive.

D) The longnose gar (Lepisosteus osseus) has a much narrower head than the devil catfish (Bagarius bagarius), and the most sensitive sensors for the devil catfish are closer to the snout than are those for the longnose gar.

Fish and Low-visibility, v.4

**Find all 4 versions of this question at https://t.me/SATakror*

Water flowing around an obstruction creates vortices (swirls) that vary in size with the obstruction's size and position, and fish sense these vortices in order to navigate around the obstructions. Using models of three fish-head shapes—narrow (low ratio of width to length), intermediate, and wide (high ratio of width to length)—Yuzo R. Yanagisuru, Otar Akanyeti, and James C. Liao showed that as vortex size increases, pressure fluctuations at the snout increase for narrow-headed and intermediate-headed fish but remain low for wide-headed fish. A second research team has therefore hypothesized that in low visibility, a fish will be more likely to avoid an obstruction when the associated pressure fluctuations at the fish's snout are greater.

89) Which finding, if true, would most directly challenge the second research team's hypothesis?

A) A study using obstructions that created large vortices in low-visibility conditions found that the Synodontis macropunctata, which has a relatively wide head, bumped into less than half of the obstructions.

B) A study using obstructions that created large vortices in low-visibility conditions found that the intermediate-headed dusky smooth-hound (Mustelus canis) bumped into the obstructions just as often as the wide-headed Synodontis macropunctata did.

C) A study using obstructions that created large vortices in low-visibility conditions found that the wide-headed Synodontis macropunctata bumped into obstructions more often than the intermediate-headed dusky smooth-hound (Mustelus canis) did.

D) A study using obstructions that created large vortices in low-visibility conditions found that some specimens of the intermediate-headed dusky smooth-hound (Mustelus canis) bumped into the obstructions more often than others did.

Household Fuel Usage, v.1

**Find all 4 versions of this question at https://t.me/SATakror*

90

In North Macedonia, use of solid fuel (e.g., coal, wood) as a share of total household fuel use fell by 25 percent between 2000 and 2019; such shifts are typically explained by appeal to the energy ladder, a model holding that fuel choice is mediated mainly by household income (specifically, high-technology fuels displace solid fuels as incomes rise). Rasmus Heltberg's study of fuel use in Nicaragua shows how reductive this model is, however: household fuel use was heterogeneous, flexible, and influenced by several factors, including local electrical and plumbing infrastructure.

90) Based on the text, which choice best explains why the author characterizes the energy ladder as "reductive"?

A) The energy ladder attributes household fuel choice primarily to a single characteristic of households, but Heltberg's study showed that multiple circumstances can affect household fuel choice.

B) The energy ladder asserts that households only consider one aspect of fuel when making fuel choices, but Heltberg's study showed that households ignore that characteristic of fuel and evaluate several other factors instead.

C) The energy ladder assumes that the same factor influences household fuel choice in all locations, but Heltberg's study showed that fuel choice in Nicaragua is influenced by different factors than is fuel choice in North Macedonia.

D) The energy ladder holds that the adoption of one type of fuel means that another type of fuel must be displaced, but Heltberg's study showed that several different fuel types are typically used in equal proportion.

Household Fuel Usage, v.2

**Find all 4 versions of this question at https://t.me/SATakror*

In Bolivia, use of solid fuel (e.g., coal, wood) as a share of total household fuel use fell by more than half between 2000 and 2015: such shifts are typically explained by appeal to the energy ladder, a model holding that fuel choice is mediated mainly by household income (specifically, high-technology fuels displace solid fuels as incomes rise). Richard Hosier and Jeffrey Dowd's study of fuel use in Zimbabwe shows how reductive this model is, however: although income of course constrained fuel choice, several factors, including the difficulty of acquiring fuel sources, influenced decisions.

91) Based on the text, the author would most likely agree with which statement about household income?

A) It can explain some but not all of the differences in fuel choice across households.

B) It is often said to influence household fuel choice but actually does not.

C) It affects household fuel choice but not for the reasons assumed by the energy ladder model.

D) It constrains the amount of fuel households use but not the type of fuel they use.

Household Fuel Usage, v.3

**Find all 4 versions of this question at https://t.me/SATakror*

In Algeria, use of solid fuel (e.g., coal, wood) as a share of total household fuel use fell by approximately three-fourths between 2000 and 2019; such shifts are often explained by appeal to the energy ladder, a model holding that fuel choice is mediated mainly by household income (specifically, high-technology fuels displace solid fuels as incomes rise). Rasmus Heltberg's study of fuel use in Ghana shows this model to be reductive, however: household fuel use was heterogeneous, flexible, and influenced by several factors, including the stability of the local electrical grid.

92) Which choice best describes the function of the information about Algeria in the text as a whole?

A) It provides an example of a type of change that the text goes on to suggest is poorly suited for evaluating whether the energy ladder is a viable model.

B) It introduces a finding that the text goes on to suggest can be explained in two different ways that are equally compelling.

C) It describes a trend that the text goes on to suggest has a similar cause as a seemingly unrelated trend observed in Ghana.

D) It illustrates the kind of phenomenon that the text goes on to suggest is frequently but inadequately accounted for by the energy ladder.

Household Fuel Usage, v.4

**Find all 4 versions of this question at https://t.me/SATakror*

In Brazil, use of solid fuel (e.g., coal, wood) as a share of total household fuel use fell by around 70 percent between 2000 and 2019; such shifts are typically explained by appeal to the energy ladder, a model holding that fuel choice is mediated mainly by household income (specifically, high-technology fuels displace solid fuels as incomes rise). Alemu Mekonnen and Gunnar Köhlin's study of fuel use in Ethiopia shows this model to be reductive, however: household fuel use was heterogeneous, flexible, and influenced by several factors, including households' cooking habits.

93) Based on the text, how would an advocate of the energy ladder model most likely explain the change observed in Brazil?

A) Between 2000 and 2019, household incomes rose in Brazil, resulting in households meeting a greater proportion of their fuel needs with high-technology fuels.

B) Beginning in 2000, high-technology fuels became more readily available in Brazil, enabling households of various income levels to shift away from solid fuels.

C) Solid fuel use declined in Brazil between 2000 and 2019 because changes in household income during that period made solid fuels more expensive relative to income.

D) Household incomes in Brazil rose from 2000 to 2019, which weakened the association between income and household fuel choice.

Bioaccumulation of nanoparticles, p.1

Some metals contain tungsten carbide nanoparticles (WC-NPs), which can leach into waterways and soils via wastewater. In a 2018 study, Mikael T. Ekvall and colleagues found that WC-NPs can accumulate in the bodies of water lice (Asellus aquaticus). While bioaccumulation of manufactured nanoparticles may be inherently worrisome, it has been hypothesized that WC-NP bioaccumulation in invertebrates like A. aquaticus could serve a valuable proxy role, obviating the need for manufacturers to conduct costly and intrusive sampling of vertebrate species – such as Perez's frogs (Pelophylax perezi), commonly used in regulatory compliance testing – for nanoparticle bioaccumulation, as environmental protection laws currently require.

94) Which finding, if true, would most directly weaken the hypothesis presented in the text?

A) Compared with P. perezi, A. aquaticus can accumulate detectable WC-NP concentrations with significantly fewer negative effects.

B) The rate of WC-NP uptake in A. aquaticus differs from the rate of WC-NP uptake in P. perezi in a way that is not yet well understood by researchers.

C) A. aquaticus has been shown to accumulate several other types of manufactured nanoparticles in addition to WC-NPs, whereas P. perezi has been shown to accumulate only WC-NPs.

D) When A. aquaticus and P. perezi are exposed to similar levels of WC-NPs, concentrations of WC-NPs in animals of both species show little variation from individual to individual.

Bioaccumulation of nanoparticles, v.2

Some pharmaceuticals contain titanium dioxide nanoparticles (*TiO2-NP*s), which can leach into waterways and soils via wastewater. In a 2014 study, Cheng Tan and Wen-Xiong Wang found that *TiO2-NP*s can accumulate in the bodies of water fleas (*Daphnia magna*). While bioaccumulation of manufactured nanoparticles may be inherently worrisome, it has been hypothesized that *TiO2-NP* bioaccumulation in invertebrates like *D. magna* could serve a valuable proxy role, obviating the need for manufacturers to conduct costly and intrusive sampling of vertebrate species—such as Chinook salmon (*Oncorhynchus tshawytscha*), commonly used in regulatory compliance testing—for nanoparticle bioaccumulation, as environmental protection laws currently require.

95) Which finding, if true, would most directly support the hypothesis presented in the text?

A) In comparable environments, *D. magna* and *O. tshawytscha* display comparable rates of *TiO2-NP* uptake.

B) Compared with *O. tshawytscha*, *D. magna* can tolerate significantly higher *TiO2-NP* concentrations without displaying any negative effects.

C) It is easier to detect low and harmless concentrations of *TiO2-NP*s in *D. magna* than it is to detect high and harmful concentrations of *TiO2-NP*s in *O. tshawytscha*.

D) *TiO2-NP* concentrations in *D. magna* tend to vary more from individual to individual than do *TiO2-NP* concentrations in *O. tshawytscha* when the species are exposed to similar levels of *TiO-NP*s.

Ectotherms and Endotherms, v.1

**Find all 3 versions of this question at https://t.me/SATakror*

96

The oceanic whitetip shark and the Pacific halibut are ectothermic (cold- blooded) fish, whereas the shortfin mako shark and the yellowfin tuna are regional endotherms—they retain metabolic heat resulting in body temperatures above the ambient water temperature. Modeling the effect of falling ambient temperatures on ectotherms indicated to researchers Haley R. Dolton and colleagues that ectotherms' body temperatures inexorably decrease toward the ambient temperature. Data from wild basking sharks show that even though their body temperature consistently remains 1.0 to 1.5°C above ambient, it also declines as the ambient temperature falls. This suggests that the body temperatures of the

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96) Which choice most logically completes the text?

A) yellowfin tuna and the shortfin mako shark are likely to be more variable than the body temperature of the basking shark.

B) Pacific halibut and the basking shark are at least partly determined by the ambient water temperature.

C) basking shark and the shortfin mako shark remain fixed even as ambient water temperature changes, while the body temperature of the Pacific halibut changes with variations in the ambient water temperature.

D) basking shark and the yellowfin tuna remain fixed even if the ambient water temperature changes, while the body temperature of the oceanic whitetip shark varies even if the ambient temperature is stable.

Ectotherms and Endotherms, v.2

**Find all 3 versions of this question at https://t.me/SATakror*

97

The blue shark and the striped marlin are ectothermic (cold-blooded) fish, whereas the salmon shark and the Pacific bluefin tuna are regional endotherms—they retain metabolic heat resulting in body temperatures above the ambient water temperature. Unlike those of ectotherms, regional endotherms' hearts have a relatively high proportion (greater than 30 %) of compact myocardial tissue, which is needed to maintain these animals' blood pressure. In a 2023 study, Haley R. Dolton and colleagues found that basking sharks—planktivorous filter-feeders that were classified as full ectotherms at the time—have hearts consisting of 47% compact myocardial tissue, thereby undermining that classification.

97) According to the text, which choice most accurately describes the importance of the percentage of compact myocardial tissue in the basking shark's heart?

A) It is insufficient to establish that the basking shark should be categorized, as Dolton and colleagues suggest.

B) It is higher than that of the salmon shark, a regional endotherm, despite the basking shark being a full ectotherm.

C) It is higher than that of the Pacific bluefin tuna, a regional endotherm, which calls into question the basking shark's status as a regional endotherm.

D) It is, by itself, sufficient to invalidate how the basking shark has been categorized in the past.

Ectotherms and Endotherms v.3

**Find all 3 versions of this question at https://t.me/SATakror*

The Pacific halibut is an ectothermic (cold-blooded) fish, whereas the yellowfin tuna is a regional endotherm; unlike that of ectotherms, its body temperature remains above the ambient water temperature. The basking shark had been classified as a full ectotherm, but researchers Haley R. Dolton et al. found that its body temperature remains 1.0 to 1.5C above ambient, which is _____ that classification.

98) Which choice completes the text with the most logical and precise word or phrase?

A) incongruous with

B) subordinate to

C) opportune for

D) indicative of

Magical Realism v.1

**Find all 3 versions of this question at https://t.me/SATakror*

Scholars cite Men of Maize, the 1949 novel by Guatemalan author Miguel Ángel Asturias, as a foundational text of magical realism, the Latin American style of fiction in which antirealistic plot devices—often borrowed from the spiritual and narrative traditions of Indigenous and colonial societies in the Americas—are deployed in an otherwise realistic mode of representation typical of the modern novel. This style has exerted a decisive influence on authors in the United States, including Susan Power, whose 1994 novel The Grass Dancer resembles classic magical realist novels in its juxtaposition of literary realism with long-established cultural traditions—namely, those of the Dakota people of the northern plains.

99) Which quotation from a literary scholar would most directly support the claim in the underlined portion of the text?

A) "The cultural traditions of the Dakota people of the northern plains, which figure so prominently in the magical realist tradition of Latin America, permit realistic as well as antirealistic scenarios—much as The Grass Dancer does."

B) "Although much of The Grass Dancer conforms to the conventions of realistic fiction, Susan Power also incorporates elements drawn from Dakota cultural traditions that transcend and expose the limitations of realism."

C) "Even though The Grass Dancer alternates between realistic and antirealistic modes of representation, the influence of Dakota cultural traditions remains constant throughout the novel."

D) "Much of the interest of The Grass Dancer derives from the productive tension between its competing influences—namely, Dakota cultural traditions and the magical realism of Latin America."

Magical Realism v.2

**Find all 3 versions of this question at https://t.me/SATakror*

100

Scholars cite Men of Maize, the 1949 novel by Guatemalan author Miguel Ángel Asturias, as a foundational text of magical realism, the Latin American style of fiction in which antirealistic plot devices—often borrowed from the folkloric traditions of Indigenous and colonial societies in the Americas—are deployed in an otherwise realistic mode of representation typical of the modern novel. This style has exerted a decisive influence on authors around the world, including Orhan Pamuk, whose 2001 novel My Name Is Red resembles classic magical realist novels in its juxtaposition of literary realism with folklore—namely, that of Turkey.

100) Which quotation from a literary scholar would most directly support the claim in the underlined portion of the text?

A) "Although Turkish folklore clearly informs the style and occasionally antirealistic plot of My Name Is Red, the novel also shows the inarguable influence of the magical realist tradition of Latin America."

B) "The logic of the realistic plot of My Name Is Red is repeatedly and productively disrupted by allegorical imagery and situations drawn from Turkish folklore."

C) "Like many works in the Latin American magical realist tradition, My Name Is Red is indebted to antirealist elements in the folkloric tradition of Turkey."

D) "While My Name Is Red alternates between realistic and antirealistic modes of representation, details suggesting the influence of Turkish folklore nevertheless occur throughout the novel."

Magical Realism v.3

**Find all 3 versions of this question at https://t.me/SATakror*

101

Magical realism, a style that weaves fantasy into realistic narratives, coalesced in Latin America in the 1960s. It is associated with Gabriel García Márquez, whose One Hundred Years of Solitude is a key text. The style was embraced by Mexican American writers, such as Ana Castillo, whose So Far from God draws on Mexican folktales as a source of fantasy. Yet The Mixquiahuala Letters, also by Castillo, features no fantastical plot events. And despite his popular reputation, García Márquez himself wrote several works of strict realism, including Chronicle of a Death Foretold.

101) Which choice best states the function of the underlined sentence?

A) To identify an unrecognized point of similarity between Chronicle of a Death Foretold and So Far from God

B) To suggest that Mexican American authors who were influenced by García Márquez may have, in turn, influenced him

C) To refute the possible impression that García Márquez conformed to magical realism in all of his works

D) To challenge the common assumption that One Hundred Years of Solitude is superior to realist works by García Márquez

Answers to SATakror
(101 most repetitive questions)

1.B	14. D	27. D	40. D	53. A	66. D	79. B	92. D
2.C	15. B	28. D	41. A	54. D	67. D	80. B	93. A
3. C	16. B	29. B	42. D	55. C	68. C	81. A	94. B
4. C	17. D	30. C	43. D	56. D	69. A	82. C	95. A
5. C	18. B	31. A	44. C	57. C	70. B	83. D	96. B
6. B	19. B	32. B	45. D	58. A	71. D	84. B	97. D
7. C	20. A	33. A	46. B	59. D	72. D	85. D	98. A
8. B	21. C	34. A	47. B	60. B	73. B	86. D	99. B
9. D	22. C	35. C	48. B	61. D	74. B	87. B	100. B
10. C	23. A	36. A	49. B	62. C	75. A	88. A	101. C
11. A	24. C	37. C	50. B	63. B	76. D	89. B	
12. B	25. B	38. D	51. A	64. B	77. D	90. A	
13. D	26. B	39. D	52. C	65. B	78. A	91. A	